NOTES DURING PHD

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CONTENTS

1. TODO. This is just a playground!

1.1. *Questions.*

- Why not we just use LASSO always if this is so good!
- Check (best write) something on *post hoc* and *ad hoc*
- What is the regularity condition in universal inference condition?
- Why do we want to have data-dependent significant level α
- What is the difference between p-process and E-process and nonnegative martingale?
- What is a adapted sequence of random sets, random variables
- What is Radon-Nikodym derivative
- What would happen if you just multiply E-variables together? Shouldn't you consider the
- What is the evidence in testing? Is likelihood ratio the best one we have for evidence
- What is the difference between carrying measure and carrying measure?
- Write down the lemma/fact that each element in a regular exponential family is continuous with respect to each other and could be used as carrier. Is it a one-to-one mapping?
- Mean-value Parameterization Convexity of mean-value spaces and canonical spaces
- Loewner ordering

1.2. *All the inequalities.* Hoeffding, Bernstein, McDiarmid, Talagrand's
Evidence lower bound
line crossing inequality average treatment effect Union bound mixture adaptive design

1.3. *Papers, talks, textbook, and more topics.* Clarke Barron 1990 1994 about

- Stein's Methods
- Ito's process
- Levy's process
- Gaussian process

1.4. *Basic Concepts.*

Definition 1.1 (Admissibility). In E-process, a process is inadmissible if there exists a process that is strictly better than it at certain time points.

Calibration
Understanding the Fourier transform in terms of transform
Placing a restriction on the Fourier transform == smaller function space
Minimax setup
Integration / Measure
Topology crash course
The topologist Stephen Smale stunned the mathematical community in 1958 [Sma58] when he proved it was possible to turn the sphere inside out without introducing any creases. Several ways to do this are beautifully illustrated in video recordings [Max77, LMM94, SFL98].
Tower properties

2. Meetings.

Recap of Theorem 1 of Francesca. https://arxiv.org/pdf/2509.01064
X follows some exponential family

$$P(X) = \frac{\exp\bigl(-\beta^T T(X)\bigr)}{Z(\beta)},$$

where β is the canonical parameter and $\mu(\beta)$ is the mean parameter in $\mathtt{M}$.
Assume n iid sample $X^{(n)} = (X_1, \ldots, X_n)$, and w a prior over the $\mathtt{M}$,

$$Q(\mu \in \mathtt{M}') = \int_{\mathtt{M}'} w(\mu)\, \mathrm{d}\mu$$

"normalized sufficient statistic converges in distribution to Q^w at rate $O(\log m/m)$."

DEFINITION 2.1. *Ineccsi* subsets Θ_0. $\mathcal{M}$ is a model with smooth parameter Θ. Please refers to the MDL book Chapter 7-10.

- the interior of Θ_0 is non-empty.
- the closure of Θ_0 is a compact subset of the interior of Θ.

THEOREM 2.2. *Let w be any (regular) prior density over mean-value space $\mathtt{M}$. For any* incessi *subset $\mathtt{M}' \subset \mathtt{M}$, we have*

$$P^w\left(\frac{T(X^{(n)})}{n} \in \mathtt{M}'\right) - Q^w\left(\mu \in \mathtt{M}'\right) = O\left(\frac{\log(n)}{n}\right)$$

The Bayesian marginal likelihood P^w is

$$P^w(X) = \int_{\mathtt{M}} P(X) w(\mu)\, \mathrm{d}\mu$$

Questions to be asked

- Regular prior density w over the mean-value space $\mathtt{M}$?
- Is the prior on mean or canonical space?
- Technque: Chernoff, Sanov property, generalized I-projection and a conditional limit theorem also in Yunda's paper

Old notes and ideas collected from meetings.

- Wouter: Betting on sure things: both null and alternative agrees on X's mean is a and null also states mean of Y is b, does it make sense to test even?
- Raf: Ideas on time consistency anytime valid comparison of algorithm/Time-consistent decision making
- Morgane Austern: Cross-validation, Consistent bound better than Chernoff bound, worse than CLT boun
- Philip Boeken: Which tests could be tested, conditional independence and so on
- Coin-betting optimality $\rightarrow$ Stochastic dominance
- alpha-spending function!
- Lars Sickert Karam (TUE): estimation for Bernoulli Trials with dependent inhibition: a surprising phenomenon @ Lunteren

20.04.2026.

Yearly evaluation.

- Why log function is natural? Everything is natural, the project between the null and the alternative becomes a likelihood ratio and it just happens that the derivative of the $\log$ is just $\frac{1}{\cdot}$ so it cancels out. But Nick has a paper saying the e-value connecting the usual NP test UMP test with the evalue. In that case, the utility function becomes the interpolation of a constant and a linear function $y = ax + b$.
- Should definitely look into the local central limit theorem!
- Projection's mixture vs mixture over projections

03.04.2026. Here I would like to make sort of the definitive changes for the figures for the e-table paper:

README.md: Font family, Font size, figsize

Asymptotic

- 4x4 plot of possible config change: GAP, the higher the worse
- 2x4 plot of stopping time distribution:
- Bayesian, NML, Turner

A asymptotic plot of catching up would be nice

Real data

- Side-by-side two plots of log-evidence plots for Trolley 1 and Trolley 2
- Gauss, Plug-in, NML, Turner, UMP, Fisher

Visual

- Parameter space visualized
- Prior one-sided and shifted are plot
- Merging and tables up and down

e-reader talk. The most important thing to be sure is that

Got confused. Got the notation, $X_1^k, \ldots, X_{n_k}^k \in \{0, 1\}$, for $n_k \geq 1$ and $k = a, b$. We denote by $Y_a = \sum_{i=1}^{n_a} X_i^a$ and $Y_b = \sum_{i=1}^{n_b} X_i^b$, essentially the sufficient statistics for n_k independent Bernoulli.

outcome	a	b	
1	$Y_{a,t}$	$Y_{b,t}$	$n_{1,t}$
0	$n_{a,t} - Y_{a,t}$	$n_{b,t} - Y_{b,t}$	$n_{0,t}$
	$n_{a,t}$	$n_{b,t}$	n_t

TABLE 1
Data B_t *at* $t \in \mathbb{N}$.

outcome	a	b	
1	Y_a^t	Y_b^t	n_1^t
0	$n_a^t - Y_a^t$	$n_b^t - Y_b^t$	n_0^t
	n_a^t	n_b^t	n^t

TABLE 2
Data B^t *until* $t \in \mathbb{N}$.

$Y_{a,t}$	$Y_{b,t}$
$n_{a,t} - Y_{a,t}$	$n_{b,t} - Y_{b,t}$

Let's just assume its just testing if there is a difference at group a and b.

Said I got the best method (e-process) at hand that collect 2-stream data, which allows me to measure the evidence whenever I want, continue/stop (optional).

Right now, its constructing a sequential conditional variable and calculate a S_t and multiply them together.

$$evidence = \prod_{i=1}^{t} S_t(Y_{a,t}, Y_{b,t}, n_{a,t}, n_{b,t})$$

I omitted and just see S_i as the *calculator* at timestep i that evaluates.
If $n_{a,t}, n_{b,t}$ is all stationary, this is all fine.

outcome	a	b
1	7	80
0	11	20

TABLE 3
A table between 50 and 51

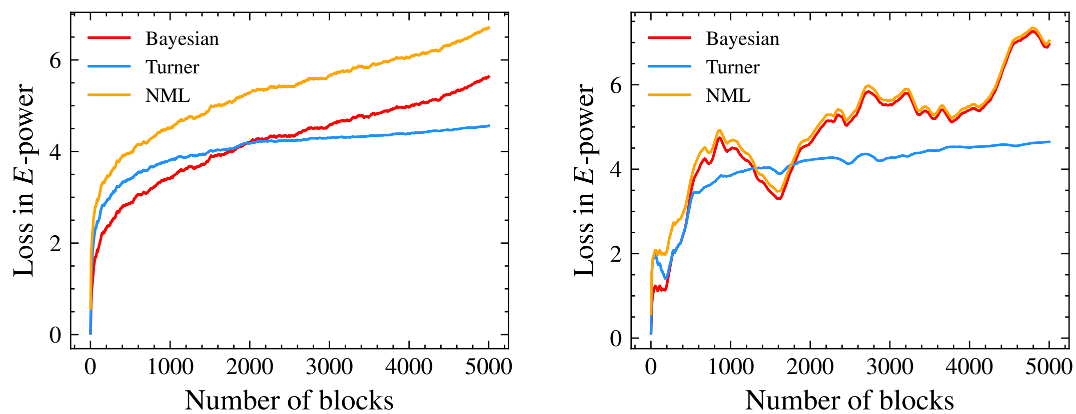
outcome	a	b
1	12	8
0	1	20

TABLE 4
A table between 100 and 101

outcome	a	b
1	$y_{a1} + \cdots$	$y_{b1} + \cdots$
0	$y_{b0} + \cdots$	$y_{b0} + \cdots$

TABLE 5
A 'growing' table

Assume the tables above we got after stop and *peak* at Table 1 Is this already the continues evaluation already?



(a) Fixed mean $(\theta_a, \theta_b) = (0.6, 0.4)$ for Bernoulli stream, group size are fixed as 5.

(b) Fixed mean $(\theta_a, \theta_b) = (0.6, 0.4)$ for Bernoulli stream, group size follows Poisson distribution with mean 5.

Fig 1

I would like to try, so I make 2 streams of data where group size is following $\max(1, \text{Poisson}(5))$, as if we collect more or less every 5 observations. stream a and b are with a fixed parameter. See Fig 1.

I suddenly thought it was I have a table (fixed) like the one above, and then altering number, but i dont think so, right? This is a different way to look at it continously I suppose.

20.03.2026. The discussion with Seb and Peter is mainly about fixing and finalizing.
Also discuss about possible venues.
We should think from the users' perspectives.

- 1 group is trying to optimize for power so use UMP each one and multiply them together they wanna publish
- 1 is trying to seq-cond without learning (essentially fixed a point mass on 1 $\delta_{\min}$)
- 1 is trying to do learning then seq bayesian (with or without updating)
- Put a prior on maybe on $(0, \infty)$ or $(\delta_{\min}, \infty)$.

12.03.2026. This week has been quite busy with hands-on stuff, mostly drawing for the ML2 project. For the python project, I am glad that things are now more organized (hopefully), but the organization or development need to stem from the User's perspective.

For example, he or she would like to just run the test ASAP given the data and the model/methods. Therefore, a ideal interaction would be

```
import MyAmazingTest as mat
data = pd.read_csv() # [10, 232]
test = mat.turner() # ump()

print(test.e_value)
```

For me personally, a good separation of what the pkg *MUST* contain or not is critical:

- IO and data validation
- calculation method, caller in FP or OOP favor both valid
- (optional) plotting
- (optional) simulation data

Stuff like plotting should not be as rigid, and it would nice to just have a proper output as standarized variables (data frame or numpy arrays). This leaves more room for optimization and tweaking later.

This is also true for simulation and large repetition. Writing 100 codes of caching function might ultimately be a waste of time. In the end, we need to think like the USER!

Discussing future math stuff with Dante.

> Thou shalt not linger in the comfort of thine data analysis!

- Asympototic and compound `E`
- Exponential conditional likelihood
- Forecasting
- `E`-closure

09.03.2026 with Peter.

- Expand a example to δ stay the same and θ_a, θ_b jump in the unit square
- Just try the method Peter for a bit, and show it this week maybe
- I should really look into whether learning, especially Bayesian learning will make the end product different, i.e. $\prod_i^n S_{\text{COND}}(Y_i) = \prod_i^n S_{\text{COND}}(Y_{\pi(i)})$ for a different permutation π
- Lots of new ideas in the list pdf!
- $d/2 \log(n)$ rate ?

Feb 2026 Work plan.

- Simulate Peter's idea: for batches $B_1, B_2, \ldots, B_n$ of size 2, a 'oracle' conditional e-variable, divided by the expectation of μ.
- R implementation of fixed delta
- *Discuss with Alex* re-do futility analysis with the 'right' e-variables (so you stop for futility if you can reject $\mathcal{H}_1$, using an e-variable that takes $\mathcal{H}_1$ to be the null hypothesis - this is in many cases not achievable by flipping the original e-variable (only do it for conditional e-variable and NML e-variable, not for p-to-e calibration because we do not know how to make p-values with $\mathcal{H}_1$ as the null) (discuss with Alexander how to treat the case $\mathcal{H}_0 : \delta = 0$ vs $\mathcal{H}_1 : \delta \neq 0$ because it's unclear how to make an e-variable for H1)

Peter's idea of optimization. We are only in the two streams case and use the same notation as the E-table paper by Sebastian.

1. This amounts to finding the avergered of μ (or δ)?

2. COND-NML divided by a sup.

For time/batches $t = 1, \dots, N$, we calculated the COND-NML E-variable for each t. This is superscript, using the data up to $t \in [N]$.

Then we got the $S^t_{\mathrm{c(NML)}}$, which is not a E-process. We obtained another sequence of pseudo E-variable $S_t = S^t_{\mathrm{c(NML)}} / S^{t-1}_{\mathrm{c(NML)}}$.

To normalize, another sequence C_t is obtained via:

$$C_t = \sup_{\mu \in [0,1]} P_\theta(ya, yb) * S^t$$

01.03.2026 Meta E. Some long discussion on if we are doing sequential analysis at all...

The z-test team is facing with a extreme outlier:

H0:

27.02.2026. Dante's talk on his work Rao-Blackwellization of E-variable. The basic idea is that conditioning on the sufficient statistics $T(X)$ can *improve*.

There're a few consequences:

- One can limit the search space to functions relating to $T(X)$ to be more efficient

He gave some examples where one can point out if E is GRO or not: exchangeable sequences.

One thing to note is bound and if the expectation holds. Log-utility function which we often use is then

Double interpretation of Bayesian methods: Average vs learning. Looking at the product for a fixed δ and then integrating (viewed as averaging over unknown parameters).

Integrating over the posterior for each block and then taking the product (viewed as learning).

Let $S(X_i \mid \delta)$ be the SEQ-COND E-variable that we consider. Assume that we do not know the oracle alternative δ^* for now but there is some ways to *learn* or *average* them.

Recall the factorization of joint distribution of $X^n = \{X_1, X_2, \dots, X_n\}$:

$$P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2 \mid X_1)P(X_3 \mid X_1, X_2) \cdots P(X_n \mid X^{n-1}) = \prod_{i=1}^{n} P(X_i \mid X^{i-1}).$$

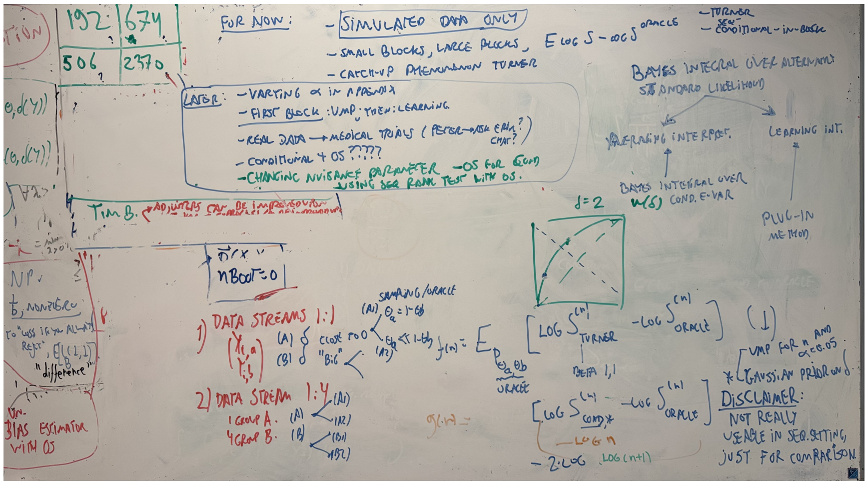
Note that this does not require *iid*!

For iid X_i:

$$\begin{aligned} P(X^n) = \prod_{i=1}^{n} P(X_i) &= \int \pi(\delta) \prod_{i=1}^{n} P(X_i)\, \mathrm{d}\delta \quad \text{Aveging} \\ &= \prod_{i=1}^{n} \int P(X_i \mid \delta)\pi(\delta \mid X^{i-1})\, \mathrm{d}\delta \quad \text{Learning} \end{aligned}$$

Note that it always holds that

$$S(X^n) = \int S(X^n \mid \delta)\pi(\delta)\, \mathrm{d}\delta = \prod_{i=1}^{n} S(X \mid \delta)$$



TODO: I would need to write this properly, especially in the misspecification case.
The misspecified case is also interesting! in 2x2 cases.

28.11.2025. *Alex*: Will collab with Oslo uni hospital and clinical trial unit. Futility Analysis): Working on re-analyzing "Many Labs" (a reproducibility project) data using E-values and futility (a stopping rule for when a study is not yielding results). He described the difficulty in mathematically quantifying the "race" between the rejection process and the futility process

The difficulty is between the planned blocks and a large practical stopping time and their relationship.

Dante CLT in KL distance and TAV and entropic CLT.

Seb Random and fixed design matrix in linear mixed effect model, full rankness of the design matrix.

27.11.2025. Talk with Peter on the 2×2 table problem again.

Paper now!. Everything is one-shot now!

- Data simulation comes a fixed oracle (θ_a^*, θ_b^*) with a δ^*. Although the COND E-variable is for a fixed δ, right now we are testing only from a point, rather than a whole curve.
- Calibrated E-value should include the version from wiki.
- No SEQ-COND in current set up for now, we do not 'learn' the alternative but rather just average them out!
- Turner's E should use a flat prior $\mathrm{Beta}(1, 1)$.
- Consider adding the $d/2 \log n$ term to avoid confusion, readers should know that this is not sequential.
- E-variable to be compared:
  - Turner's
  - Oracle (just multiple) this is the SEQ-RIPr
  - COND (Not SEQ!): NML, Gaussian
  - UMP

Paper future!.

- SEQ-COND
- misspecification case: SEQ-COND and COND is different by a fraction $\frac{\mathrm{Var}(R)}{\mathrm{Var}(\theta^*)}$.

delta in General case paper is fixed. GRO(W) is only discussed in right haar case!

Yunda's paper discussion about the δ or β, the canonical parameters of FNCH is always fixed. Therefore, the prior W_1 or W_0 on the alternative hypothesis is always w.r.t. a particular fixed δ.

21.11.2025. *Dante* Rant about the KL divergence and its disconnection from the sample space, why not Wasserstein distance which is very popular in optimal transport.

Peter E-processes and Approximations. T-test and coarser filtrations $X_i - X_1$. I guess he also talked about the exp. family.

for exponential families where the null and alternative share the same sufficient statistic, conditioning on the sufficient statistic yields an E-variable (expectation 1).

Problem: This process is generally not an E-process.

Comparison: It performs better than Universal Inference (UI) but lacks the E-process property.

Prequential Plug-in NML. Assume parametric model and simple null and alternative hypothesis. We observe iid data stream $X^t = \{X_1, X_2, \dots, X_t\}$. To make it even simpler, the support of X: $\mathcal{X}$ is finite.

$$\begin{aligned}\mathcal{H}_0 &: X \sim P(X \mid \theta_0)\\ \mathcal{H}_1 &: X \sim P(X \mid \theta^*)\end{aligned}$$

Now let's look at the plug-in, for the first observation X_1.

$$S^1_{\text{NML}} = \frac{P(X_1 \mid \hat{\theta}(X_1))}{P(X_1 \mid \theta_0) \sum_{k\in\mathcal{X}} P(k \mid \hat{\theta}(k))},$$

where $\hat{\theta}(\cdot)$ is the MLE estimator.

Check this is a E-variable:

$$\mathbb{E}_0\left[S^1_{\text{NML}}\right] = \sum_{X\in\mathcal{X}} P(X \mid \theta_0) \cdot \frac{P(X \mid \hat{\theta}(X))}{P(X \mid \theta_0)\sum_{k\in\mathcal{X}} P(k \mid \hat{\theta}(k))} = \sum_{X\in\mathcal{X}} \frac{P(X \mid \hat{\theta}(X))}{\sum_{k\in\mathcal{X}} P(k \mid \hat{\theta}(k))} \leq 1.$$

For the second time point, we shift the X_2 over $\mathcal{X}$ and obtained a MLE estimator:

$$S^2_{\text{NML}} = \frac{P(X_2 \mid \hat{\theta}(X_1, X_2))}{P(X_2 \mid \theta_0) \sum_{k\in\mathcal{X}} P(k \mid \hat{\theta}(X_1, k))}$$

Going to time t

$$S^t_{\text{NML}} = \frac{P(X_t \mid \hat{\theta}(X_1, X_2, \dots, X_t))}{P(X_t \mid \theta_0) \sum_{k\in\mathcal{X}} P(k \mid \hat{\theta}(X_1, X_2, \dots, k))} = \frac{P(X_t \mid \hat{\theta}(X^t))}{P(X_t \mid \theta_0) \sum_{k\in\mathcal{X}} P(k \mid \hat{\theta}(\tilde{X}^t))}$$

Given the canonical filtration $\mathcal{F}_{t-1}$,

$$\mathbb{E}_0\left[S^t_{\text{NML}} \mid \mathcal{F}_{t-1}\right] = \sum_{\mathcal{X}} \frac{P(X_t \mid \hat{\theta}(X_1, X_2, \dots, X_{t-1}, X_t))}{\sum_{k\in\mathcal{X}} P(k \mid \hat{\theta}(X_1, X_2, \dots, X_{t-1}, k))}$$

Noted that this is a form of SEQ-COND so expect a lot worse than COND (log linearly). Going back to the FNCH distribution, the support of Y_a is luckily finite: $\Gamma = [\gamma_{\min}, \gamma_{\max}]$. Similarly, we observed iid stream data $Y^t = \{Y_1, Y_2, \dots, Y_t\}$, where $Y_i = (Y_{i,a}, Y_{i,b})^\intercal$ are the counts of ones in group a, b. Numbers of ones in both group is denoted by $n_{i,1} = Y_{a,i} + Y_{b,i}$.

The sum up to time t is denoted by $n_1^t = \sum_{i=1}^t n_{1,i} = \sum_{i=1}^t Y_{a,i} + Y_{b,i}$. As we assume group size is stationary, $n_a^t = n_a \cdot t, n_b^t = n_b \cdot t$. We can write the NML conditional E-variable for Y^1:

$$S^1_{\mathrm{NML}} = \frac{f_{\mathrm{FN}(n_a,n_b,n_{1,1},\hat{\delta}_1)}(Y_{a,1})}{f_{\mathrm{FN}(n_a,n_b,n_{1,1},0)}(Y_{a,1}) \sum_{k \in \Gamma} f_{\mathrm{FN}(n_a,n_b,n_{1,1},\hat{\delta}_{1*})}(k)}, \quad \hat{\delta}_1 = \arg\max_{\mathbb{R}} f_{\mathrm{FN}(n_a,n_b,n_{1,1},\delta)}(Y_{a,1})$$

For the second time point, the MLE estimator is based on Y^2 and the normalizing MLE in the denominator is based on shifting all possible values.

$$S^2_{\mathrm{NML}} = \frac{f_{\mathrm{FN}(n_a,n_b,n_{2,1},\hat{\delta}_2)}(Y_{a,2})}{f_{\mathrm{FN}(n_a,n_b,n_{2,1},0)}(Y_{a,2}) \sum f_{\mathrm{FN}(n_a,n_b,n_{2,1},\hat{\delta}_{2*})}(k)}.$$

$$\hat{\delta}_2 = \arg\max_{\mathbb{R}} f_{\mathrm{FN}(n_a^2,n_b^2,n_1^2,\delta)}(Y_{1,a} + Y_{2,a})$$

$$\hat{\delta}_2 * = \arg\max_{\mathbb{R}} f_{\mathrm{FN}(n_a^2,n_b^2,n_1^2,\delta)}(Y_{1,a} + k),$$

where $k \in [\max(0, n_{2,1} - n_b), \min(n_a, n_{2,1})]$.

14.11.2025. Jelle presented his paper on Multiple testing in e-value https://arxiv.org/abs/2509.02517.

The main issue/difficulty is how does one overcome the large combinatorial number 2^N where N is the number of hypothesis tested: By identifying the "worst-case" subset for a given size, the complexity drops from exponential to polynomial.

You can also add DAG in hypothesis class: e.g. if 1 holds, 3 must hold.

Post-hoc: the set of hypothesis can be chosen post-hoc! e.g., choosing a specific brain region) while maintaining valid error control.

He compared with e-BH, Benjamini-Yekutieli and Su's. By reformulating existing methods through the E-closure principle, they could achieve uniform improvements (more rejections for the same error control).

09.11.2025 Talk with Alex about Bayesian. Assume a single parametric family p with prior π. Similarly we consider a stream of data $X^n = \{X_1, X_2, \ldots, X_n\}$.

In time step 1:

$$\begin{aligned} p(X^1) = p(X_1) &= \int_\Theta p(X_1 \mid \theta)\pi(\theta)\, \mathrm{d}\theta \\ &= p\left(X_1 \mid \theta\right) \end{aligned}$$

At time step 2:

$$\begin{aligned} p(X^2) = p(X_1, X_2) &= \int_\Theta p(X_1 \mid \theta) p(X_2 \mid \theta)\pi(\theta)\, \mathrm{d}\theta \\ &= p(X_1)p(X_2 \mid X_1) = \int_\Theta p(X_1 \mid \theta)\pi(\theta)\, \mathrm{d}\theta \int_\Theta p(X_2 \mid \theta)\pi(\theta \mid X_1)\, \mathrm{d}\theta \end{aligned}$$

07.11.2025. Mostly just Lunteren conference. Sebastian Engelke's Talk: Sebastian Engelke gave a talk on predicting floods in Bern using generalized Pareto distributions and conditional models based on upstream measurements (rain, river flow).

Stephan mention some interesting examples on the Bernoulli example!, Also there is a interesting bernoulli Fisher information gain example in Lunteren!

Lukas presented the BAM problem for shifting means: Goal is to find a algorithm to identify best arm (unique)

Algorithm is called importance Sampling for Shifting Means.

Abstract: We study an interactive testing environment in which the learner chooses one of K distributions to sample each timestep with the goal of finding the distribution with the highest expected mean with high probability in the least number of overall samples (the best-arm identification setting with known confidence). We coin the Shifting Means setting, a non-stationary setting where an offset is added to all samples which can change fully adversarially between timesteps and demonstrate how the known approaches of the Track-and-Stop algorithm and the Generalized Likelihood Ratio Test (GLRT) fail for Shifting Means. We propose generalized Importance Weights for Shifting Means, which we show to be delta-correct and sample efficient. We also present a lower bound and evaluate our results empirically.

10.10.2025. *Wouter*: He mentioned his work in ALT paper, Lucas will give a talk in 2 weeks. Lucas will stay here for another year

Dante: He showed to Seb that E-variables/process is the same for random and fixed design. Also talked about something on the covariate in connecting to Nick's paper.

Stephan: RL, server cluster at ML

Peter: There is small issues with safestats packages in terms of Leiden's group,

Alex: subGaussian: The main challenge is determining the conditions under which a distribution remains subGaussian after conditioning, without simply assuming it is "conditionally subGaussian".

07.10.2025. Let's assume we are talking about a classification model that predicts if a image is a cat or a dog or a human.

Conformal Prediction vs. Bayesian Methods.

- Bayesian posterior probabilities give number in $(0, 1)$ to 3 labels that sums to 1.
- A key difference is that conformal scores represent *upper bounds* on probability and don't need to sum to one, which contrasts with standard probability theory.
- e.g., the conformal on the cat or the dogs can be both 20%, sort of the upper bound, while the human is 80%, then it does not sum up to 1.
- Bayesian can be overconfident because of a prior, even a non-informative one, introduces some strong assumptions.
- Is there a real 'FLAT' prior that provides no information?
- Conformal prediction can be seen as a "worst-case" approach, whereas Bayesian methods are "best-case." Conformal may sometimes be too pessimistic.

Distinction Between Research Areas.

- Sebastian is on *model evaluation* and *calibration*, which is more aligned with Bayesian statistics. His goal is to find the best overall predictor or model from a set of candidates, e.g., which weather forecasters is the best among 100.
- But conformal prediction focuses on the *individual prediction* at each step, providing a set of possible outcomes with a certain confidence level (e.g., 95%), without trying to determine the best underlying model.
- I am still not sure about how one formulate the conformal prediction: Is it a sup over the training space?

Challenges with E-variables.

- A major practical issue with sequential e-variables is that the cumulative product can be drastically reduced if multiplied by a single small e-value. This causes significant fluctuations.
- A simple "safeguard" is proposed: mixing the e-variable with a constant, such as using $0.5 + 0.5 \times E$. This ensures the multiplier is always at least 0.5, preventing a total loss of evidence.
- This is caused *damping* and it's very popular in fin math where going to infinity is often not realistic and too theoretical!
- Using *Rényi e-variables* is suggested as a potentially more sophisticated solution to this volatility problem.

Paper Writing and Next Steps.

- The professor advises starting the writing process with a clear "story" and an outline, saving the polished introduction for the end.
- The main story for their current paper is extending existing methods for 2x2 tables to handle more realistic scenarios where data arrives in a single, potentially asymmetric batch rather than in sequential pairs.
- The immediate next step is to conduct an experiment using "Rosanna's e-variable." This involves breaking their large data table into many small, sequential blocks, as this method performs best with small group sizes.

Potential Future Research Topics.

- *Stein's Method*: A technique for proving the central limit theorem that, unlike traditional methods, provides explicit error bounds for finite samples.
- *Prior-Marginal Asymptotic Symmetry*: A long-standing research question about the relationship where the marginal distribution of a sufficient statistic tends to mimic the shape of the prior distribution on the parameter (mean-value space). Stein's method might be a useful tool for developing a direct proof.
- Show the above phenomenon in exponential case (Bernoulli), which has already been verified.

26.09.2025. There are quite a lot of things that I need to check after this. Nick presented on conformal prediction where my take-away about is that one could relate the

Another theme is still the contingency table. There is something strange about the NML sequential in which the estimation is on the entire sequence.

Another problem that I need to write down is that for a given prior π and Bayesian updating on the blocks, is the multiplication equal to the global one.

Simpler proof: 1) write something in closed form, e.g. exponential 2) compare it to the proof of seq-RIPr and RIPr in general case. 3) but I think the seq-COND is not there yet?

I need to refresh on NP-lemma, product measure, outer measure, push-forward measure and etc.

I also wanna do a step-by-step guide on plotting and figure in a standalone repo.

05.09.2025.

What I did this week. In terms of coding, what I did is transform the big genotype G and phenotype P data into a small subset. There are mainly two reasons: 1) 455k SNPs exceeds our scope of multiple testing, and we really need a handful of interesting targets 2) the 44 SNPs in the small dataset are not "significant"

1) Run Fisher exact test with `PLINK -assoc fisher` 2) Rank them by p-value, FDR control (0.05) 3) Obtain about 125 SNPs with very small p

Noted that this is for the 4688 sample points where we are simply testing with Fisher exact test for association. Furthermore, I synthesise sequential dataset by shuffling into blocks of 50 whose ratio of case between control is close to global.

	SNP+ (a)	SNP- (b)
case (1)	1779	866
ctrl (0)	4089	2626

One might think that in a global significant SNPs, the pseudo-sequential data blocks would likely yield significant results. However, we observed multiple $p_i = 1$, e.g., on tables as below.

	a	b
1	46	29
0	16	9

Furthermore, we are trying to calculate the probability of $p_{\text{fisher}} = 1$ for the pseudo blocks. Start with the simple one and use union bound.

$$\mathrm{P}\left(\frac{n'_{a1}}{n'_{a0}} \approx \frac{n_1}{n_0} \,\middle|\, n'_a, n'_b, \frac{n_a}{n_b} \approx \frac{n'_a}{n'_b}\right)$$

For now, we consider the cases, $\approx$ is $=$ avoiding situations where non-integers appears and ceiling/flooring functions are applied.

- Check out this https://scholar.google.com/citations?user=UnrY-40AAAAJ&hl=en
- https://en.wikipedia.org/wiki/Boole%27s_inequality

29.08.2025. I mainly talked about the gap between the test supermartingale and E-process. I was confused but Peter corrected me, saying even though you can find a supermartingale upper bounding the E-process.

In the simple null case, they are the same. However, in the composite settings, it is not, for each $P \in \mathcal{P}$, there is a test supermartingale but there is not a single one for all $P \in \mathcal{P}$. In fact, it's very easy to give examples where a random process is E-process but not supermartingale.

He mentioned examples from UI, which is very straightforward to show it's Eand Alex also said something about the canonical filtrations, by Ramdas, in the t-test.

Seb talks about Gaussian channelling, a paper on 2004 with 3rd derivative and information theory. Interesting, worth a look.

25.08.2025. Practical goal:

- Compare speed of conditional E in BiasedUrn and R: about 3x performance, not worth it to import
- Add save E into the package, not done yet but the styles are already similar to the safestats pacage

This week's goal theory-wise

- Just write down again the KL for Gaussian family for repetition.
- Restate the connection between integral and sequential product of UI: haven't found the literature yet
- Try to re-state the simple and anti-simple case
- What is the seq-RIPr and seq-COND?
- Why do we need to have a general KL measure in general paper
- Read carefully how the maximum is found in the log-optimal paper
- Reproduce with safestats packages for the
- UMP in math. stat. course lecture notes.
- I wanna write down $\mathrm{d}\nu$ and $\mathrm{d}X$ and all that.

The difference between the simple and anti-simple case In short, the simple case is where $\Sigma_q - \Sigma_p$ is negative which mean we can find a RIPr via a prior (or a element) of P.

15.08.2025. Sebastian did a great presentation regarding testing quantile given filtrations. I would formulate here and also add a picture.

The question in mind is to test whether data X is from a hypothesis $\mathcal{H}$ that is non-parametric:

$$\mathcal{H} = \{P \in \mathcal{P}(X) \mid \mathbb{E}_p\left[\phi_i(X) = 0\right], i = 1, 2, \ldots, d\}$$

where $\mathcal{P}(X)$ is all distribution on X.

The simplest instance where testing X with the same mean μ where $\phi_1(X) = X - \mu$. Larson has proven that the 'optimal' E-variable must be in the form:

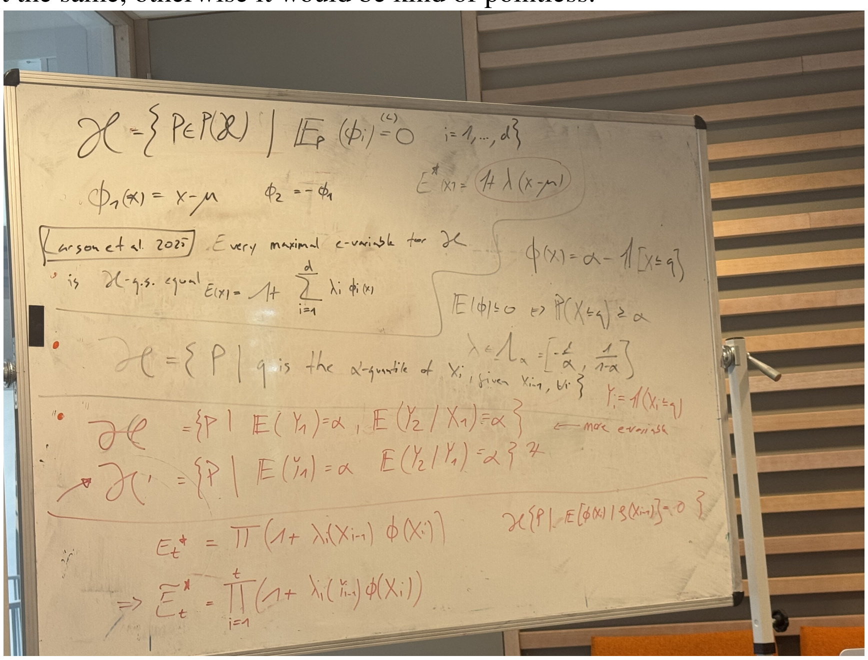
$$S := 1 + \sum_{i=1}^{d} \lambda_i \phi_i(X)$$

Sebastian is mainly interested in the cases if there is any gain in E-variables compared to a coraser filtrations. Imagine we have conditioned on the original data $X_i, i = 1, 2, \ldots,$ you would think that the hypothesis case constructed as below:

$$\mathcal{H} = \{P \in \mathcal{P}(X) \mid \mathbb{E}[Y_2 \mid X_1] = \alpha\} \tag{2.1}$$

$$\mathcal{H}' = \{P \in \mathcal{P}(X) \mid \mathbb{E}[Y_2 \mid Y_1] = \alpha\} \tag{2.2}$$

where $Y_i = 1_{X_i \leq q}$. He showed that both $\mathcal{H}$ and $\mathcal{H}'$ are convex and $\mathcal{H}$'s closed convex hull is not the same, otherwise it would be kind of pointless.



14.08.2025. This is a short discussion regarding UI, specifically about the difference between prequential and integral representation.

Supposed iid data $\mathcal{D} = \{X_1, X_2, \ldots, X_n\}$ and we denote the data up to time i by $X^{(i)} = \{X_1, X_2 \ldots, X_i\}$.

One way to instantiate UI by

$$\frac{\prod_{i=1}^{n} P_{\tilde{\theta}_{alt|i-1}}(X_i)}{P_{\hat{\theta}_0}(X^{(n)})}$$

where $\tilde{\theta}_{alt|i-1}$ is any estimator in alternative based on the first $i-1$ data points $X^{(i-1)}$ and $\hat{\theta}_0 = \arg\max_{\theta \in \Theta_0} \prod_i p_\theta(X_i)$ is the MLE estimator under the null. See more details in section 7 of UI paper.

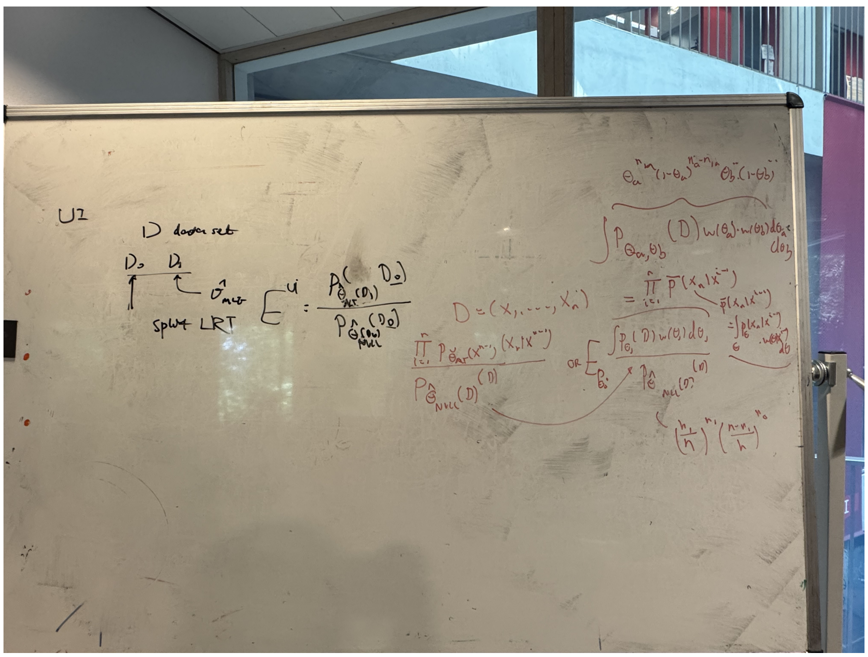
Notice that the sequence of the data $\mathcal{D} = \{X_1, \ldots X_n\}$ really matters. Imagine you obtian $\mathcal{D}'$ with a rearranged sequence and thus slightly different $\tilde{\theta}$ and hence slightly different value at the denominator.

Revisit factorization of probability, it's also called chain rule or general product rule:

$$
\begin{aligned}
\mathrm{P}\left(X_1, X_2, \ldots, X_n\right) &= \mathrm{P}\left(X_1\right) \mathrm{P}\left(X_2 \mid X_1\right) \cdots \mathrm{P}\left(X_n \mid X_1, X_2, \ldots, X_{n-1}\right) \\
&= \prod_{i=1}^{n} \mathrm{P}\left(X_i \,\middle|\, X^{(i-1)}\right)
\end{aligned}
$$

Question 2.3. Below is a mixture of all nonnegative test martingale/e-process?

$$
\frac{\int p_\theta(X^{(n)}) w(\theta)\, \mathrm{d}\theta}{P_{\hat{\theta}_{\text{null}}}(X^{(n)})}
$$



25.07.2025. I only briefly went over the conformal prediction and fisher's noncentral hypergeometric distribution. There were some discussions on what the conformal prediction is.

Imagine you have a classifier for images (dogs, cat, etc.) and is trained via N datasets. For the next prediction, we need something to quantify the uncententy to say that

> The label for X_{N+1} I gave being X (here could be any label), has

Peter gave a algorithmic explanantion where the predicted labels are gaven for each label, then run through against the previous training datasets. Rank them, cut off the tailing α percent then we can say we are confident about our prediction with $1 - \alpha$.

However, this is awfully similar to the permutation test, by Sebastian. Yeah it does look a lot like just ranking the prediction and give a p-value.

Alexander also suggested using Gaussian for conformal prediction might be too confusing as the parameter and the prediction kind of just are the same thing. Maybe try Poisson example where parameter is in real number while the prediction is in $\mathbb{Z}$.

What I do not follow is the output of the classifier is a weighted matrix over all the labels. What is the ranking being done over? Is it

21.05.2025. Papers discussed: PNAS and general case

For the simple case (the Gaussian location family in 3.2.1 in **?**), my interpretation is as followed: If negative semidefnite $\Sigma_q - \Sigma_p$, then we reduced to the simple case where the RIPr E-variable is not only in $\mathrm{conv}(\mathcal{P})$ (the convex hull of the null dist.) but rather a element of $\mathcal{P}$ itself.

Otherwise, we can find a prior on $\mathcal{P}$ with sharp variance $\mathcal{N}(\boldsymbol{\mu}^*, (\Sigma_q - \Sigma_p)/n)$. The results in turn suggests that we cast a sharp prior on $\mathcal{P}$ with the same mean. However, extra argument about the alternative Q, I think we move to the distributions from Q that shares the same sufficient statistics. This is still the part where I am having some doubts. I am still sure is that what does it mean to operate on a 'enlarged' or 'modified' alternative even if the alternative $\mathcal{Q} = \{Q\}$ being just Gaussian with same mean and a different variance.

Another point is in the Anti-Simple case, then $S^{(n)}_{Q,\text{SEQ,RIP}} = 1$. First of all, the superscript $^{(n)}$ says it is considering a sequential settings but $X^{(i)}$ is *singular* following either null dist. or alternative dist. This $S = 1$ breaks simply when we are considering two X following the same distribution, we then test whether $X' = (X, Y) \overset{\text{iid}}{\sim} P \in \mathcal{P}$ or alternative.

Data spilting in UI In a sense, the spilting is done sequentially or done across n data points in contrast to the D_0 D_1 in the original paper. Now we are talking about the UI in (3.2.14) I think! The classical on would be $L = L_0(\hat{\theta}_1)/L_0(\hat{\theta}_0)$

$$S^{(n)}_{\breve{\boldsymbol{\mu}},\text{UI}} = \frac{\prod_{i=1}^{n} q_{\breve{\boldsymbol{\mu}}|_{i-1}(U_{(i)})}}{p_{\breve{\boldsymbol{\mu}}|_n(U^{(n)})}}$$

We have a regular ML estimator likelihood in the denominator after looking at the whole sequence n. For the nominator, we have a product of the ML likelihood for each time i. In the end, we have effectively losing just $U_{(1)}$ in calculating likelihood.

Beyond NP We discussed about the similarities among the loss function in the ERM scheme and in this paper. The ERM loss is always with respect to some data (empirical) while here we are concerning about a data-dependent loss?

The problem or rather common difficulty with p-value in NP paradigm is that why do we report p-value at all? What does it mean for a small p-value when $p \ll \alpha$?

Peter proposed a alternative as in we should report E-value instead. I think also the notation of seeing α as the error probability is somewhat challenged.

Inspiration In vaccination study, we see a extremely small p-value on null but we are not allowed to make *new* decision without setting up new hypothesises and new studies. In other words, if we looked at a promising data *post-hoc*, there is really not much we can do based on the original data. But with E-value, he argued that new decision can be based/conditioned on the data.

Is l in Eq. 1 **?** just a new α?

In this loss function $L(\kappa, a)$, κ is the true state of nature that we don't know? We just assumed if the data is coming from null or alternative.

In all, I think the main idea is to change from Type-I error safe to Type-I risk safe. I think section 2's main take home message is that any maximally compatible decision rule must be a E-variable

This is cited from Micheal I Jordan's notes: https://people.eecs.berkeley.edu/~jordan/courses/260-spring10/other-readings/chapter8.pdf

3. TLDR for papers. Here I really hope that I can somehow summarise in my words of the talks or presentation I have been to or literature I have read into a *short* paragraph.

There are only two requirement:

- short: ideally in one sentence, but it should definitely shorter than the abstract
- memorable/impressionable: for example, instead of precisely stating the LLN, it is better to write "keep tossing a fair coin, we are sure about the average being 0.5".

Admissible Anytime-Valid Sequential Inference Must Rely on Nonnegative Martingale (**?**):

> p-process, e-process, stopping times are sort of the same: we can always find a nonnegative martingale for them. However, there is a gap.

The numeraire e-variable and reverse information projection (**?**):

> Shows that there is always a numeraire E-variable X^* in the form of $\mathbb{E}_Q\left[X/X^*\right] \leq 1$ for every E-variable X in the simple alternative Q vs composite null $\mathcal{P}$ setting. It is unique (up to Q-nullset), log-optimal, and connection between the effective null !

Reverse Information Projections and Optimal E-statistics (**?**):

> There is always a RIPr (?) even when the KL is infinite in the simple alternative Q vs composite null $\mathcal{P}$ setting. In simple words, if the KL is infinite (bad at describing compared), there is still a relatively better version to describe that!.

	a	b
1	46	29
0	16	9

4. Fisher noncentral hypergeometric distribution. For a 2×2 table as shown above, n_a and n_b denote the group sizes. Let y_a and y_b be the number of successes in groups a and b, respectively. The total number of ones is $n_1 = y_a + y_b$.

Conditional on the margins n_a, n_b, n_1, y_a follows Fisher's noncentral hypergeometric distribution.

$$y_a \mid n_a, n_b, n_1, \delta \sim \mathrm{FNCH}(n_a, n_b, n_1, \delta).$$

Replacing y_a with x

$$f(x \mid \delta, n_a, n_b, n_1) = \frac{\binom{n_a}{x}\binom{n_b}{n_1-x}e^{\delta x}}{\sum_{\gamma^-}^{\gamma^+}\binom{n_a}{k}\binom{n_b}{n_1-k}e^{\delta k}} = h(x)\exp\{\delta \cdot T(x) - A(\delta)\}. \tag{4.1}$$

This is a 1-dimensional exponential family wrt δ,

- domain: $[\gamma^-, \gamma^+]$, $\gamma^- = \max\{0, n_1 - n_b\}$ and $\gamma^+ = \min\{n_1, n_a\}$
- sufficient statistics: $T(x) = x$
- natural parameter: δ
- cumulant function $A(\delta; n_a, n_b, n_1) = \log Z(\delta; n_a, n_b, n_1)$
- partition function $Z(\delta; n_a, n_b, n_1) = \sum_{\gamma^-}^{\gamma^+}\binom{n_a}{k}\binom{n_b}{n_1-k}e^{\delta k} = \sum_{\gamma^-}^{\gamma^+} g(n_a, n_b, n_1, k)e^{\delta k}$ where $g(n_a, n_b, n_1, k) = \binom{n_a}{k}\binom{n_b}{n_1-k}$

When $\delta = 0$, it reduces to central hypergeometric distribution,

$$f(x \mid 0, n_a, n_b, n_1) = \frac{\binom{n_a}{x}\binom{n_b}{n_1-x}}{\sum_{\gamma^-}^{\gamma^+}\binom{n_a}{k}\binom{n_b}{n_1-k}} = \frac{\binom{n_a}{x}\binom{n_b}{n_1-x}}{\binom{n_a+n_b}{n_1}} \tag{4.2}$$

4.1. *MLE estimation.* The log-likelihood function for FNCH is

$$\ell(\delta) = \log\binom{n_a}{x} + \log\binom{n_b}{n_1 - x} + x\delta - \log\left(\sum_k \binom{n_a}{k}\binom{n_b}{n_1-k}\exp(\delta k)\right) \tag{4.3}$$

$$= \log h(x) + \delta \cdot x - \log Z \tag{4.4}$$

Differentiate the log likelihood function l wrt δ and setting it to zero:

$$\frac{\partial \ell(\delta)}{\partial \delta} = x - \frac{\sum_{\gamma^-}^{\gamma^+} k\binom{n_a}{k}\binom{n_b}{n_1-k}e^{\delta k}}{\sum_{\gamma^-}^{\gamma^+}\binom{n_a}{k}\binom{n_b}{n_1-k}e^{\delta k}} = 0 \tag{4.5}$$

This is the same as finding δ such that

$$X = \mathbb{E}[X] \tag{4.6}$$

4.2. *Software references.*

- R: `BiasedUrn::dFNCHypergeo(x = ya, m1 = na, m2 = nb, n = n1, odds = odds)`, where odds=$\exp(\delta)$. Here x is the number of 1s in group a, m_1, m_2 are the group sizes, and n is the total number of 1s. https://www.rdocumentation.org/packages/BiasedUrn/versions/1.07/topics/BiasedUrn-Multivariate
- Python: `scipy.stats.nchypergeom_fisher(k=ya, M=na+nb, n=na, N=n1, odds)`. https://docs.scipy.org/doc/scipy/tutorial/stats/discrete_nchypergeom_fisher.html

5. Conditional E-variables for 2x2 tables.

5.1. *Definitions and calculation methods.*

5.1.1. *Conditional E-variable.* The FNCH distributional setup is collected in Section 4. Conditioning on the total number of ones, and using the pdf from Eq. 4.1,

$$S(x \mid \delta, n_a, n_b, n_1) = \frac{f(x \mid \delta, n_a, n_b, n_1)}{f(x \mid 0, n_a, n_b, n_1)} \tag{5.1}$$

$$= \exp\{\delta x + A(0; n_a, n_b, n_1) - A(\delta; n_a, n_b, n_1)\} \tag{5.2}$$

As we do not know δ, we marginalized wrt some prior on δ, which is the most computational intensive part.

$$S(x \mid n_a, n_b, n_1) = \int_{\mathbb{R}} S(x \mid \delta, n_a, n_b, n_1) p(\delta) \, \mathrm{d}\delta$$

5.1.2. *Two Bernoulli random variables.* *Heavily adapted on Dante's Thesis.*
Let X, Y be two independent Bernoulli random variables,

$$X \sim \mathrm{Bernoulli}(\theta_a) \quad Y \sim \mathrm{Bernoulli}(\theta_b).$$

$\mathcal{P}$ describes the set of distributions on (X, Y) where X, Y are independent and marginally distributed as Bernoulli. A distribution on (X, Y) could be parameterized by P_{θ_a, θ_b} where θ_a, θ_b ranges in $[0, 1]$. There's a *natural* way to re-parameterize the tuple via log-odds ratio,

$$\delta := \log \frac{\theta_a}{1 - \theta_b} \Big/ \frac{\theta_b}{1 - \theta_b} \in \mathbb{R}.$$

Note that two parameters (a tuple) are still needed, the log-odds ratio δ is chosen such that we can inexplicitly represent a composite set, i.e.

$$P_\delta = \{P_{\theta_a, \theta_b} \mid \log\left(\frac{\theta_a}{1 - \theta_b} \cdot \frac{1 - \theta_b}{\theta_b}\right) = \delta \in \mathbb{R}\}.$$

Assuming point null and alternative hypothesis, it is easy to formulate the testing problem by a likelihood ratio test,

$$\mathcal{H}_0 : X, Y \sim P_{\theta_a, \theta_b} \quad \mathcal{H}_1 : X, Y \sim P_{\theta_a^*, \theta_b^*}.$$

The problem extends easily to sequential test in iid settings.

$$\mathcal{H}_0 : (X_1, Y_1), (X_2, Y_2), \ldots \overset{\text{iid}}{\sim} P_{\theta_a, \theta_b} \quad \mathcal{H}_1 : (X_1, Y_1), (X_2, Y_2), \ldots \overset{\text{iid}}{\sim} P_{\theta_a^*, \theta_b^*}.$$

5.1.3. *Setup and notation.* We denote the Kullback–Leibler divergence by $\mathrm{D}_{\mathrm{KL}}(P \,\|\, Q) = \int \log(\,\mathrm{d}P/\mathrm{d}Q\,)\,\mathrm{d}P$ where $\mathrm{d}P/\mathrm{d}Q$ is the Radon-Nikodym derivative of P with respect to Q and the integral is taken on the same probability space (not shown here). We will assume that the measures discussed here all have full support.

We start with a binary random variable U following Bernoulli distribution with parameter μ, here μ is also the mean in mean-value parameterization. The density reads

$$p_\mu(U) := \mu^U \cdot (1 - \mu)^{1-U}, \quad U \in \{0, 1\}, \quad \mu \in [0, 1]$$

By the standard properties of regular exponential families, we parameterize $\mathcal{P}$ in terms of the mean-value parameter space $\mathtt{M}_p$ and write

$$\mathcal{P} = \{P_\mu : \mu \in \mathtt{M}_p\}, \quad \mathtt{M}_p = [0, 1].$$

The canonical parameterization for any $\mu \in \mathtt{M}_p$ is defined as

$$p_\beta(U) = \frac{1}{Z_p(\beta)} \cdot \exp(\beta \cdot U) \cdot p_\mu(U),$$

where $Z_p(\beta)$ is the normalizing constance and the canonical parameter space is defined as $\mathtt{B}_p = \{\beta : Z_p(\beta) < \infty\} = \mathbb{R}$.

Moving on to the 2-stream examples with different group size n_a, n_b. At each time point i, n_a and n_b copies of Bernoulli variables are collected, following the same distribution:

$$U_a = (U_{a,1}, U_{a,2}, \cdots, U_{a,n_a})^T \sim \mathrm{Ber}(\boldsymbol{\mu}_a)$$
$$U_b = (U_{b,1}, U_{b,2}, \cdots, U_{b,n_b})^T \sim \mathrm{Ber}(\boldsymbol{\mu}_b)$$

The sufficient statistics vector is denoted by $X = (\sum_j^{n_a} U_{a,j}, \sum_j^{n_b} U_{b,j})$. $\mathcal{P}$ is extended by the i.i.d. assumption to a set of distributions for random process $U_{(1)}, U_{(2)}, \ldots$ for $i \geq 1$, i.i.d. copies of $U_{(i)} = (U_{(i),a}, U_{(i),b})$. Using superscript, we denote the random process up to time i by $U^{(n)} := (U_{(1)}, U_{(2)}, \ldots U_{(n)})$ and $X^{(n)} := (X_{(1)}, X_{(2)}, \ldots X_{(n)})$ for the sufficient statistics respectively.

The *alternative* we consider first $\mathcal{Q} = \{Q\}$ is simple with some oracle $\boldsymbol{\mu}^* = (\boldsymbol{\mu}_a^*, \boldsymbol{\mu}_b^*)$.

$$\mathcal{H}_1 : U_a \sim \mathrm{Ber}(\boldsymbol{\mu}_a^*), U_b \sim \mathrm{Ber}(\boldsymbol{\mu}_b^*) \quad \text{independent.}$$

Under mean-value parameterization:

$$\mathcal{H}_1 : X_a \sim P_{\boldsymbol{\mu}_a^*}, X_b \sim P_{\boldsymbol{\mu}_b^*} \quad \text{independent.}$$

The null distribution $\mathcal{P}$ is the set of distributions where group a and b are equipped with the same parameter and hence composite:

$$\mathcal{H}_0 : X_a \sim P, X_b \sim P \quad \text{i.i.d. for some } P.$$

5.1.4. *Convexity prerequisite.*

Remark 5.1. The Bernoulli model, $\mathcal{P} = \{P_\theta : \theta \in [0,1]\}$, on $\mathcal{Y} = \{0,1\}$ is convex, whereas the product model $\mathcal{P}^n = \{P_\theta^n : \theta \in [0,1]\}$ on $\mathcal{Y}^n$ is not convex for $n > 1$.

Proof. Convexity is straightforward for single observation case $Y = \{0,1\}$. For $\lambda \in [0,1]$, consider the convex combination of two elements, $P_{\theta_1}, P_{\theta_2} \in \mathcal{P}$. Probability for $\lambda P_{\theta_1} + (1-\lambda)P_{\theta_2}$,

$$\mathrm{P}(Y=1) = \lambda\theta_1 + (1-\lambda)\theta_2$$
$$\mathrm{P}(Y=0) = 1 - \big(\lambda\theta_1 + (1-\lambda)\theta_2\big)$$

Therefore, the mixture is again in $\mathcal{P}$ with parameter $\theta^\circ = \lambda\theta_1 + (1-\lambda)\theta_2 \in [0,1]$.

Take $n = 2$, consider the mixture

$$P = \frac{1}{2}P_0^2 + \frac{1}{2}P_1^2 \tag{5.3}$$

where P_0^2 put all its mass on $(0,0)$ and P_1^2 put all its mass on $(1,1)$.

However, there is no $\theta \in [0,1]$ for P_θ^2 such that,

$$P_\theta^2(0,0) = P(0,0) = \frac{1}{2} \qquad P_\theta^2(1,1) = P(1,1) = \frac{1}{2}$$
$$P_\theta^2(1,0) = P(1,0) = 0 \qquad P_\theta^2(0,1) = P(0,1) = 0$$

$\square$

Why finite convexity does not immediately imply closure under arbitrary mixtures. Let

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}.$$

Convexity of $\mathcal{P}$ means that for every finite collection

$$P_{\theta_1}, \ldots, P_{\theta_k} \in \mathcal{P}$$

and weights

$$\lambda_1, \ldots, \lambda_k \geq 0, \qquad \sum_{i=1}^{k} \lambda_i = 1,$$

we have

$$\sum_{i=1}^{k} \lambda_i P_{\theta_i} \in \mathcal{P}.$$

However, an arbitrary mixture has the form

$$P_W = \int P_\theta \, dW(\theta),$$

where W may have infinite support.

Convexity alone directly handles finite sums such as

$$\sum_{i=1}^{k} \lambda_i P_{\theta_i}.$$

It does not directly imply that

$$\int P_\theta \, dW(\theta) \in \{P_\theta : \theta \in \Theta\}.$$

A simple example showing the issue is the open interval

$$\mathcal{C} = (0, 1) \subset \mathbb{R}.$$

The set $\mathcal{C}$ is convex. Every finite convex combination of points in $\mathcal{C}$ remains in $\mathcal{C}$. However, convexity alone does not imply that limits of such convex combinations must remain in $\mathcal{C}$.

For example, take

$$x_m = \frac{1}{m}.$$

Then $x_m \in \mathcal{C}$ for every m, but

$$\lim_{m \to \infty} x_m = 0.$$

The limit point satisfies

$$0 \notin \mathcal{C}.$$

Thus convexity alone does not imply closedness under limits or arbitrary infinite averaging. Therefore, to conclude that an arbitrary mixture belongs to the model, one needs an additional argument.

In the finite sample-space case, this additional argument is supplied by Carathéodory's theorem.

Why the integral is in the convex hull.

Assume the sample space Y is finite:

$$Y = \{y_1, \ldots, y_m\}, m \in \mathbb{N}.$$

Then each distribution P_θ can be represented by the vector

$$p_\theta = \big(p_\theta(y_1), \ldots, p_\theta(y_m)\big) \in \mathbb{R}^m.$$

The mixture distribution P_W has probability mass function

$$p_W(y) = \int p_\theta(y) \, \mathrm{d}W(\theta).$$

Therefore its vector representation is

$$p_W = (p_W(y_1), \ldots, p_W(y_m)).$$

Substituting the definition of $p_W(y_i)$ gives

$$p_W = \left(\int p_\theta(y_1)\, dW(\theta), \ldots, \int p_\theta(y_m)\, dW(\theta)\right) \tag{5.4}$$

$$= \int \big(p_\theta(y_1), \ldots, p_\theta(y_m)\big)\, dW(\theta) \tag{5.5}$$

$$= \int p_\theta \, dW(\theta). \tag{5.6}$$

This is an average, or barycenter, of the points

$$\{p_\theta : \theta \in \Theta\} \subset \mathbb{R}^m.$$

Therefore

$$p_W \in \operatorname{conv}\{p_\theta : \theta \in \Theta\}.$$

Since this convex hull is finite-dimensional, Carathéodory's theorem implies that there exist

$$\theta_1, \ldots, \theta_k \in \Theta$$

and weights

$$\lambda_1, \ldots, \lambda_k \geq 0, \qquad \sum_{i=1}^{k} \lambda_i = 1,$$

with $k \leq m + 1$, such that

$$p_W = \sum_{i=1}^{k} \lambda_i p_{\theta_i}.$$

Thus the arbitrary mixture p_W can be represented as a finite mixture.

If the model $\{P_\theta : \theta \in \Theta\}$ is convex, then this finite mixture is again equal to some single model distribution:

$$\sum_{i=1}^{k} \lambda_i P_{\theta_i} = P_{\theta^\circ}$$

for some $\theta^\circ \in \Theta$.

Therefore

$$P_W = P_{\theta^\circ}.$$

5.1.5. *Bayesian mixture update.* Assume the data generating process means are fixed, group sizes are also fixed, we constructed a e-process if we *hack* and know the oracle log-odds ratio, at time t,

$$S(B_t) = S_{\text{COND},\delta}(B_t) = \frac{p_{\text{FN}}}{p_{\text{HYP}}}$$

where p_{FN} is the Fisher's noncentral hypergeometric mass and p_{HYP} is the central hypergeometric mass.

Cumulative product is worse linearly compared to GRO when sample size increase,

$$S^n_{\text{GRO}} - \prod_1^n S_{\text{COND},\delta}(B_t) \approx O(n)$$

while the 'oneshot' COND is approximately GRO

$$S^n_{\text{GRO}} - S_{\text{COND},\delta}(B^n) \approx O(1)$$

Since we do not know δ, we try a bayesian mixture

$$S(B_t) = S_{\text{COND},W}(B_t) = \int S_{\text{COND},\delta}\, \mathrm{d}W\,.$$

Let's say we do not update W, always use the same W, cumulative evidence is

$$\prod_{i=1}^t \int S_{\text{COND},\delta}\, \mathrm{d}W\,.$$

Can this be simplified?

But that's boring, we need to update the prior density,

$$\int S_{\text{COND},\delta}\, \mathrm{d}W_{t-1}\,.$$

How do I update this

5.1.6. *Bayesian prior calculations.*

5.1.6.1. Integral over prior.

5.1.6.2. Gaussian prior. Suppose we choose a standard Gaussian prior on $\mathbb{R}$

$$\delta \sim \mathcal{N}(\mu_0, \sigma_0^2).$$

After observing one table, with conditional observation $Y_a = y_a$, the posterior density is

$$p(\delta \mid y_a, n_a, n_b, n_1) \propto \exp\left\{\delta y_a - A(\delta) - \frac{(\delta-\mu_0)^2}{2\sigma_0^2}\right\}.$$

This posterior is generally not Gaussian and does not have a standard closed form.

For multiple tables, suppose we have observed $B_i = (y_{a,i}, n_{a,i}, n_{b,i}, n_{1,i})$, $i = 1, \ldots, t$

$$p(\delta \mid B^t) \propto \exp\left\{\delta \sum_{i=1}^t y_{a,i} - \sum_{i=1}^t A_i(\delta) - \frac{(\delta-\mu_o)^2}{2\sigma_0^2}\right\}$$

5.1.6.3. Conjugate prior. In the literature, conjugate prior for general exponential family can be written:

$$p(\delta) \propto \exp\{n_0 x_0 \delta - n_0 A(\delta)\}.$$

where x_0 is suggested to use the expectation of x under $\delta = 0$, which is hypergeometric distribution so $x_0 = \frac{y_a \cdot y_b}{n}$.

After seeing one data point, assume n_a, n_b, n_1 fixed, the posterior is

$$p(\delta \mid y_a, n_0, x_0) \propto \exp\left\{(n_0 x_0 + y_a)\delta - (n_0 + 1)A(\delta)\right\}.$$

It should be noted that the conjugacy only works for same margin, else the cumulant function would change.

then what is the point of conjugate then?

The *conditional* **E**-*value* with a fixed δ is

$$S_{n_a,n_b,n_1,\delta}(y_a) = \frac{f_{\mathrm{FN}(n_a,n_b,n_1,\delta)}(y_a)}{f_{\mathrm{FN}(n_a,n_b,n_1,0)}(y_a)}.$$

The 'Bayes' is w.r.t. a (truncated) Gaussian with zero mean and scale δ,

$$\int_{\mathbb{R}} S_{n_a,n_b,n_1,\delta}(y_a) \cdot \pi(\delta)\, \mathrm{d}\delta$$

See: https://github.com/wyq977/e-table/blob/yongqi/src/e_table/methods/bayes.py

5.1.7. *IID and convexity.* In the original testing framework, both null and the alternative require sample to be distributed iid, i.e., For every *binary random variable* X in group a or group b, at time $j \in \mathbb{N}$,

$$X_{a,1}^{(j)}, X_{a,2}^{(j)}, \ldots, X_{a,n_a}^{(j)} \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_a).$$

We observe two streams of binary random variables

$$X_{a,i}^{(j)}, X_{b,i}^{(j)} \in \{0,1\},$$

where

- $g \in \{a,b\}$ denotes the group,
- $j \in \mathbb{N}$ denotes the batch/time index,
- i denotes the within-batch observation index.

It is also true that at the coarse filtrations (batch level), X also following the same Bernoulli.

That is to say, for stream a, we are always tossing the same coin over and over again in time, and we just observe them in the coarse filtration of n_a coins.

The finest filtration is alway two infinite streams of Bernoulli, i.e., two coin-flipping experiments.

1. Now, null is tossing the same coin in group a and b, alternative is tossing two different coin in group a and b. But the coins remains the same throughout. We collect them at a coarse filtration, fixed, $n = n_a + n_b$.
2. Still the same null and alternative, could I change the collections (coarsening)?, i.e., every time it's not guaranteed n_a, n_b in group a and b. I could have stopped and observed $5, 10$ in the first time, and observed $100, 20$ in the second time.
3. Assuming we have the same coarsening, i.e., we observed same n_a, n_b throughout. Within each observation block, we use the same coin (iid), but between blocks, we changed coins.
4. I am not sure how to formulate this one, maybe the null is fixed two coins and the alternative is anything but, coarsening variable.

5.1.8. *Turner e-process calculation.* **Based on following assumptions**

- Group size are fixed as n_a, n_b, $n = n_a + n_b$
- At time j, $X_{a,1}^{(j)}, X_{a,2}^{(j)}, \ldots, X_{a,n_a}^{(j)}$ are n_a binary random variables
- We usually only consider the sum $Y_{a,j} = \sum_{i=1}^{n_a} X_{a,i}^{(j)}$
- Use superscript to denote the accumulated sum, $Y_a^{j-1} = \sum_{t=1}^{j-1} Y_{a,t}$, i.e., the total number of ones in group a up to time $j-1$, likewise for n_a^{j-1}

Write the e-variable with a prior W at time j

$$S_{W,j} := \frac{\prod_{i=1}^{n_a} \int_\Theta p_{\theta_a}(X_{a,i}^{(j)})\,\mathrm{d}W(\theta_a)}{\prod_{i=1}^{n_a}(\frac{n_a}{n}\int_\Theta p_{\theta_a}(X_{a,i}^{(j)})\,\mathrm{d}W(\theta_a) + \frac{n_b}{n}\int_\Theta p_{\theta_a}(X_{a,i}^{(j)})\,\mathrm{d}W(\theta_b))} \cdot \frac{\prod_{i=1}^{n_b} \int_\Theta p_{\theta_b}(X_{b,i}^{(j)})\,\mathrm{d}W(\theta_b)}{\prod_{i=1}^{n_b}(\frac{n_a}{n}\int_\Theta p_{\theta_a}(X_{b,i}^{(j)})\,\mathrm{d}W(\theta_a) + \frac{n_b}{n}\int_\Theta p_{\theta_a}(X_{b,i}^{(j)})\,\mathrm{d}W(\theta_b))}$$

If we update the prior along the way,

$$S_{\text{TUR},j} := \frac{\prod_{i=1}^{n_a} \int_\Theta p_{\theta_a}(X_{a,i}^{(j)})\,\mathrm{d}W_{|_{j-1}}(\theta_a)}{\prod_{i=1}^{n_a}(\frac{n_a}{n}\int_\Theta p_{\theta_a}(X_{a,i}^{(j)})\,\mathrm{d}W_{|_{j-1}}(\theta_a) + \frac{n_b}{n}\int_\Theta p_{\theta_a}(X_{a,i}^{(j)})\,\mathrm{d}W_{|_{j-1}}(\theta_b))} \cdot \frac{\prod_{i=1}^{n_b} \int_\Theta p_{\theta_b}(X_{b,i}^{(j)})\,\mathrm{d}W_{|_{j-1}}(\theta_b)}{\prod_{i=1}^{n_b}(\frac{n_a}{n}\int_\Theta p_{\theta_a}(X_{b,i}^{(j)})\,\mathrm{d}W_{|_{j-1}}(\theta_a) + \frac{n_b}{n}\int_\Theta p_{\theta_a}(X_{b,i}^{(j)})\,\mathrm{d}W_{|_{j-1}}(\theta_b))}$$

As we choose the W such that θ_a, θ_b are marginally independent Beta distributed, and the conjugality between Beta and Bernoulli, the posterior simplifies,

$$\int_\Theta p_{\theta_a}(X_{a,i}^{(j)})\,\mathrm{d}W_{|_{j-1}}(\theta_a) = p_{\breve{\theta}_{a|_{j-1}}}(X_{a,i}^{(j)}).$$

$$q_a = \breve{\theta}_{a|_{j-1}} = \mathbb{E}_{W_{|_{j-1}}}[\theta_a] = \frac{Y_a^{j-1} + \alpha_a}{n_a^{j-1} + \alpha_a + \beta_a}$$

$$q_b = \breve{\theta}_{b|_{j-1}} = \mathbb{E}_{W_{|_{j-1}}}[\theta_b] = \frac{Y_b^{j-1} + \alpha_b}{n_b^{j-1} + \alpha_b + \beta_b}$$

$$q_0 = \breve{\theta}_{0|_{j-1}} = \frac{n_a}{n_a + n_b} q_a + \frac{n_b}{n_a + n_b} q_b$$

write more general case where W is any prior on alternative parameter space Θ_1 instead ofx

So the Turner e-variable evaluated at time j reads

$$\begin{aligned}
S_{\text{TUR},j} :=& \frac{\prod_{i=1}^{n_a} p_{q_a}(X_{a,i}^{(j)})}{\prod_{i=1}^{n_a}(\frac{n_a}{n}p_{q_a}(X_{a,i}^{(j)}) + \frac{n_b}{n}p_{q_b}(X_{a,i}^{(j)}))} \cdot \frac{\prod_{i=1}^{n_b} p_{q_b}(X_{b,i}^{(j)})}{\prod_{i=1}^{n_b}(\frac{n_a}{n}p_{q_a}(X_{b,i}^{(j)}) + \frac{n_b}{n}p_{q_b}(X_{b,i}^{(j)}))} \\
=& \frac{\prod_{i=1}^{n_a} p_{q_a}(X_{a,i}^{(j)})}{\prod_{i=1}^{n_a} p_{q_0}(X_{a,i}^{(j)})} \cdot \frac{\prod_{i=1}^{n_b} p_{q_b}(X_{b,i}^{(j)})}{\prod_{i=1}^{n_b} p_{q_0}(X_{b,i}^{(j)})} \\
=& \frac{P_{q_a}(Y_{a,j})}{P_{q_0}(Y_{a,j})} \cdot \frac{P_{q_b}(Y_{b,j})}{P_{q_0}(Y_{b,j})} \quad \text{only consider the sum, binomial}
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}_P\left[S_{\text{TUR},j} \mid \mathcal{F}_{j-1}\right] &:= \mathbb{E}_P\left[\frac{P_{q_a}(Y_{a,j})}{P_{q_0}(Y_{a,j})} \cdot \frac{P_{q_b}(Y_{b,j})}{P_{q_0}(Y_{b,j})} \,\middle|\, \mathcal{F}_{j-1}\right] \\
&:= \mathbb{E}_P\left[\frac{P_{q_a}(Y_{a,j})}{P_{q_0}(Y_{a,j})} \,\middle|\, \mathcal{F}_{j-1}\right] \cdot \mathbb{E}_P\left[\frac{P_{q_b}(Y_{b,j})}{P_{q_0}(Y_{b,j})} \,\middle|\, \mathcal{F}_{j-1}\right] \quad \text{independence under null}
\end{aligned}$$

Just focusing on group a, we replace $\breve{\theta}_a\mid_{j-1}, \breve{\theta}_0\mid_{j-1}$ by q_a, q_0

$$\begin{aligned}
&\mathbb{E}_P\left[\frac{P_{q_a}(Y_{a,j})}{P_{q_0}(Y_{a,j})} \,\middle|\, \mathcal{F}_{j-1}\right] \\
&= \sum_{y=0}^{n_a} \binom{n_a}{y} p^y (1-p)^{n_a-y} \frac{q_a^y (1-q_a)^{n_a-y}}{q_0^y (1-q_0)^{n_a-y}} \\
&= \sum_{y=0}^{n_a} \binom{n_a}{y} \left(p\frac{q_a}{q_0}\right)^y \left((1-p)\frac{1-q_a}{1-q_0}\right)^{n_a-y} \\
&= \left(p\frac{q_a}{q_0} + (1-p)\frac{1-q_a}{1-q_0}\right)^{n_a} = C_a^{n_a} \quad \text{binomial theorem.}
\end{aligned}$$

Term C_a can be further simplified

$$\begin{aligned}
p\frac{q_a}{q_0} + (1-p)\frac{1-q_a}{1-q_0} &= \frac{pq_a \cancel{- pq_aq_0} + q_0 - q_0p - q_0q_a + \cancel{q_0pq_a}}{q_0(1-q_0)} \\
&= \frac{pq_a + q_0 - q_0p - q_0q_a}{q_0(1-q_0)} \\
&= \frac{pq_a + q_0 - q_0p - q_0q_a - q_0(1-q_0) + q_0(1-q_0)}{q_0(1-q_0)} \\
&= 1 + \frac{pq_a - q_0p - q_0q_a + q_0^2}{q_0(1-q_0)} \\
&= 1 + \frac{p(q_a-q_0) - q_0(q_a-q_0)}{q_0(1-q_0)} \\
&= 1 + \frac{(p-q_0)(q_a-q_0)}{q_0(1-q_0)}.
\end{aligned}$$

So the conditional expectation simplifies to

$$\left(1 + \frac{(p-q_0)(q_a-q_0)}{q_0(1-q_0)}\right)^{n_a} \left(1 + \frac{(p-q_0)(q_b-q_0)}{q_0(1-q_0)}\right)^{n_b} = (1+c(q_a-q_0))^{n_a}(1+c(q_b-q_0))^{n_b},$$

where

$$c := \frac{p-q_0}{q_0(1-q_0)}.$$

Taking logs,

$$\log \mathbb{E}_P\left[S_{\text{TUR},j} \mid \mathcal{F}_{j-1}\right] = n_a \log(1 + c(q_a-q_0)) + n_b \log(1 + c(q_b-q_0)).$$

By the concavity of the function $f(x) = \log(1+cx)$,

$$\begin{aligned}
&\frac{n_a}{n_a+n_b}\log(1+c(q_a-q_0)) + \frac{n_b}{n_a+n_b}\log(1+c(q_b-q_0)) \\
&\le \log\left(1 + c\left[\frac{n_a}{n_a+n_b}(q_a-q_0) + \frac{n_b}{n_a+n_b}(q_b-q_0)\right]\right) = \log 1 = 0.
\end{aligned}$$

5.1.9. *E-zoo comparison.*

5.1.9.1. UI. Based on the general case paper and the UI slide.

Let $\hat{\mu}_{|i-1}$ be any *non-anticipating*

What does it mean?

estimator based on the first $i-1$ observations and we denote the MLE estimator for the null as $\hat{\mu}_{|n} = \arg\max_{\mu \in \Theta_0} p_\mu(U^{(n)})$.

$$S_{\text{UI}} := \frac{\prod_{i=1}^{n} q_{\hat{\mu}_{|i-1}}(U_{(i)})}{p_{\hat{\mu}_{|n}}(U^{(n)})}$$

By i.i.d. assumption, the denominator reads the likelihood of $U^{(n)}$ given the MLE estimator. It is mentioned that *prequential ML* method is used here.

5.1.9.2. RIPr. By Theorem 1 in Turner and Grünwald (2023a), the GRO $\mathtt{E}$-variable reads:

$$S_{\text{RIPR}} := \frac{p_{\mu^*}(U^{(n)})}{p_{\mu^\circ}(U^{(n)})},$$

where the prior put mass 1 on the boundary and $\mu^\circ = (n_a \mu_a^*/n + n_b \mu_b^*/n)$.

REMARK 5.2. In 2-stream Bernoulli and the null hypothesis is 2 group is distributed with the same distribution $P \in \mathcal{P}$, $S_{\text{COND}} = S_{\text{RIPR}} = S_{\text{PSEUDO}} = S_{\text{MIX}}$, which achieves GRO by corollary 1 of safe testing paper.

5.1.9.3. NML. Also look at the luckiness

5.1.9.4. COND. We can first write the conditional $\mathtt{E}$-variable for a data block $X = U = (U_a, U_b)$:

$$S_{\text{COND}} := \frac{q(X_a, X_b \mid X_a + X_b)}{p(X_a, X_b \mid X_a + X_b)} = \frac{q(X_a \mid X_a + X_b)}{p(X_a \mid X_a + X_b)} = \frac{f_{\text{FN}(n,n_a,Z,w)}(X_a)}{f_{\text{hyp}(n,n_a,Z)}(X_a)}.$$

REMARK 5.3. Right now, I finally realize 4.2.3 of Yunda's thesis where the $\mathtt{E}$is:

$$S_{\text{COND}} := \frac{p_{\boldsymbol{\mu}}(X^{k-1} \mid Z)}{p_{\langle \mu \rangle}(X^{k-1} \mid Z)}.$$

Here the $Z := \sum_{j=1}^{k} X_j$ is the sum of the sufficient statistics and X^{k-1} would be the sufficient statistics losing 1 *degree of freedom*.

In our special case of Bernoulli streams, the conditional distribution corresponds to the Fisher's noncentral hypergeometric (FN) distribution and the denominator is simply the cases when odds ratio $\omega = 1$ (hypergeometric distribution).

$$S_{\text{COND}} = \prod_{i=1}^{n} \frac{q(X_a^{(i)} \mid Z^{(i)})}{p(X_a^{(i)} \mid Z^{(i)})} = \prod_{i=1}^{n} \frac{f_{\text{FN}(n,n_a,Z^{(i)},w)}(X_a^{(i)})}{f_{\text{hyp}(n,n_a,Z^{(i)})}(X_a^{(i)})}$$

5.1.10. *Calibrator.* First we run the standard Fisher's exact test and then obtain the $\mathtt{E}$-variable via a calibrator $E = 1/\sqrt{q} - 1$.

5.2. *Questions I already answered.*

5.2.1. *Prior or turning distribution.* We assume the oracle is some fixed point μ^* Let π be the prior distribution on the alternative distributions. The prior predictive distribution is defined as

$$p(U) = \int p_\mu(U)\pi(\mu)\,\mathrm{d}\mu\,.$$

In the discussion with Ly on 02.05.2025, he pointed many names for the above expression: maximal likelihood distribution, the prior predictive distribution (Bayesian), and Bayes mixture (MDL).

Additionally, he mentioned that we usually used LETTERS (X, A) for observables where GREEK LETTERS are reserved for unknown (π, θ, Θ), especially in Bayesian framework.

In contrast to the belief's μ in Bayesian, MDL principle regards π on μ rather as a turning tool, uninformative, to approach the oracle point μ^* given incoming data.

5.2.2. *What is effect size and why do we need it?.* Turner, Ly and Grünwald (2024) deals with effect size by stating that $\mathcal{H}_1$ consists of hypotheses with a minimal effect size δ while the parameter space of interest is $\mathcal{H}_0(\delta)$ (null hypothesis spaces) in Turner and Grünwald (2023a). These has confused me a great deal but I will try to explain in my words:

The effect size is the minimal difference between group a and group b we would like to see when rejecting the null hypothesis. Since we are concerned with a parametric model of Bernoulli stream in both group, parameterized by θ_a and θ_a. The effect size would essentially be the divergence d between θ_a and θ_b in the parameter space.

5.2.3. *Why put a prior on the boundary only.* W_1^*: Prior on $\Theta(\delta)$; W_1': Prior over $\Theta_1^>(\delta)$. What are good reasons of putting all prior mass on the boundary?

Pros:

- Computationally simpler as we only calculate once on θ_a and we get $\theta_b = \theta_a + \delta$ for free.
- If (θ_a^*, θ_b^*) sits somewhere near the line, E-variable growth rate is higher with larger prior support.
- Theorem 2 in Turner and Grünwald (2023a) guarantees a degenerate prior on the boundary $\Theta(\delta)$

Cons:

- If $d_{abs}(\theta_a^*, \theta_b^*) \gg \delta$, E-value calculated from W_1' is much larger than W_1^*

~~What does it mean to *pick $\mathcal{H}_1$ restricted to $\Theta(\delta)$*~~?

In the discussion with Peter, he mentioned that the project P^* might not be on the diagonal line unlike section **??**. It might just be a highly non-convex function that we can only solve by finding the min. of KL divergence after discretization. He mentioned something about the projection not being straight but rather curve.

I think I made a mistake here, when looking at the divergent case, the null would be the divergence bigger than δ while the alternative would be else?

can we plug in $\theta_b^\circ = \theta_a^\circ + \delta$ here? If so, it would just be solving a cubic equation of θ_a°.

5.2.4. *Extended null and one-shot conditional e-variable.* The extended null hypothesis is meaningful now!

In this setup that I call "one-shot" where the whole data U_a, U_b is seen at once, the conditional E-variable is optimal.

The extension of the null hypothesis is meaningful, in this parametric Bernoulli family only: the extended null $\mathcal{H}_0'$ will be all distributions where all permutation U are the same and the marginal distributions of all (all exchangeable sequences).

In other words, let $U' = (U_1, U_2, \ldots, U_{n_a+n_b})$ and π be any permutation operator on $\{1, \ldots, n_a + n_b\}$. The null hypothesis is now extended to a much bigger one, namely, whether two streams U_a, U_b are exchangeable,

$$\mathcal{H}_0' := (U_1, U_2, \ldots, U_{n_a+n_b}) \stackrel{d}{=} U_{\pi(1)}, U_{\pi(2)}, \ldots, U_{\pi(n_a+n_b)}\}.$$

Moreover, the marginal distribution of $U_j \stackrel{d}{=} U_i$ for any i, j but the conditional dependence is not assumed in $\mathcal{H}_0'$ any more in contrast to independence assumed in the original $\mathcal{H}_0$ where $U_i, U_j \stackrel{\text{iid}}{\sim} P$.

Nevertheless, the S_{COND} is still GRO-optimal, we can claim it is a *robust* E-variable for a bigger problem.

5.2.5. *Sequential-ness and conditional-ness.* I was confused about the formulation between the sequential ratio and the (what I called) global conditional E-variables.

Let X, Y be Bernoulli variables with p and q, then we can write the conditional likelihood ratio test (Wald's) as follow:

$$R^n := \prod_{i=1}^{n} \frac{f_1(X_i, Y_i \mid X_i + Y_i)}{f_0(X_i, Y_i \mid X_i + Y_i)}$$

and $f_1(\cdot) := (1+\theta)^{-1}$ and $f_0 = (1+\theta^{-1})^{-1}$

Here is the example of composite testing:

$$\mathcal{H}_0 : (X_1, Y_1), (X_2, Y_2), \ldots \stackrel{\text{iid}}{\sim} P \in \mathcal{P}_1 \text{ v.s.} \mathcal{H}_1 : (X_1, Y_1), (X_2, Y_2), \ldots \stackrel{\text{iid}}{\sim} P \in \mathcal{P}_\theta$$

where $\mathcal{P}_1$ and $\mathcal{P}_\theta$ are distribution parametrized by odds ratio θ.

Now it is almost the same as before but just extended where each data point (X_1, Y_1) is replaced by a contingency table. I think the reason I got so confused is here: we are testing m copies of tables T_i and within that T_i, each table is a 2x2 table of n_a, n_b Bernoulli streams with p, q.

Using the notation $X_i^b, i = 1, \ldots, n_a$ Ber and $X_i^a, i = 1, \ldots, n_b$ The full expression reads:

$$\frac{f_1(X_1^a, X_2^a, \ldots X_{n_a}^a, X_1^b, X_2^b, \ldots X_{n_b}^b \mid Y_a + Y_b)}{f_0(X_1^a, X_2^a, \ldots X_{n_a}^a, X_1^b, X_2^b, \ldots X_{n_b}^b \mid Y_a + Y_b)}$$

For a particular table T_i, I can just jump to the common doc, the hypothesis being tested is

For all the 2x2 tables T_i, group a and group b frequency has no difference.

$$\mathcal{H}_0 : T_1, T_2, \ldots \stackrel{\text{iid}}{\sim} P \in \mathcal{P}_1 \quad \text{v.s.} \quad \mathcal{H}_1 : T_1, T_2, \ldots \stackrel{\text{iid}}{\sim} P \in \mathcal{P}_\theta$$

Questions to discuss tomorrow:

- Is the E-process-ness still here? *YES*, by
- Is the optional continuation still here, I think so
- For the mixture, θ gets Bayesian updating each time (numerical)
- I think we assume group size n_a, n_b is stationary right? Does the
- I remember there is some nice properties concerning the concave of the Fisher's information

5.3. *Questions without answers.*

5.3.1. *Conditioning, margins, and coarsening.*

This is based on the talk with Wouter after e-reader.

Start simple, assume two streams of binary random variable X, Y indexed by t. If they are both iid Bernoulli, then we are in the paired example where each time we threw two coins. At each time, X_t or Y_t is observed, which is governed by some process or switch. For simplicity we also assume this process to be Bernoulli iid with parameter $\eta > 0$, i.e. $p(I_t = 1) = \eta$ (seeing a action from group a). It assumed independent for simplicity now and note that the means are stationary and iid.

$$P(X_t \mid I_t) \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_a)$$

$$P(Y_t \mid I_t) \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_b)$$

	a	b
1	0	1
0	0	0

Let the process goes on for n rounds, and we collected the data until time n into the summarized table.

	a	b	
1	n_{a1}	n_{b1}	n_1
0	n_{a0}	n_{b0}	
	n_a	n_b	n

In plain english, we are interested in the presence of the differences between two streams. $\eta = 1$ corresponds to the paired-sample settings where we *must* observe two binary variables (two coins). In other words there's one more degree of freedom by allowing $\eta \in (0, 1)$.

- $\mathcal{H}_0 : X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_0)$ and $Y_1, Y_2, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_0)$
- $\mathcal{H}_1 : X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_a)$ and $Y_1, Y_2, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_b)$

Let δ be some measure of divergence between θ_a and θ_b. Conditioning on n_a, n_b, n_1 gives the FNCH distribution from Eq. 4.1.

Instead, we can just condition on the total number of ones

$$P(n_{a1} \mid n_a + n_b, n_1, \delta) =$$

The questions remains are:

- Without knowledge on specific n_a, n_b, we can still formulate this into a likelihood ratio test between δ and $\delta = 0$, but I think this is composite null vs composite alternative. This is easy to see, let $r = \frac{n_a}{n_b} \in (0, 1)$, the likelihood ratio with the above formulas must contain r. And r could take value in $(0, 1)$, making the null and alternative composite.
- Is it natural to also condition on n_a on top of $n = n_a + n_b, n_1$?

A quick word on the Group Invariance paper, we are testing Gaussian with any variance.

$$\mathcal{H}_0 : X \sim \mathcal{N}(0, \sigma) \quad \sigma \in \mathbb{R}$$
$$\mathcal{H}_1 : X \sim \mathcal{N}(\mu, \sigma), \quad \mu, \sigma \in \mathbb{R}.$$

The variance is a nuisance parameter that testing simple does not care. In other words, we can normalized or scale the data however we like. The question is: the conditioning on n_1 is in the same spirit and we implicitly were also doing the same conditioning n_a (and n) so that the composite null *collapse* into a single distribution.

5.3.2. *Is MIX also an e-process?.* Adopting the same notation as in Yunda's thesis,

5.3.3. *Why Turner's e-variable for the first table is 1.*

5.3.4. *Finding confidence sequence for case 1.* First of all, can we find CS for this testing setup. If so, the confidence set should be searched through the Θ_0 (dashed red line in Fig. **??**) or the entire $[0, 1]^2$?

5.3.5. *Convexity of hypothesis spaces.* I am not sure why the modified settings where Turner and Grünwald (2023a) and Turner, Ly and Grünwald (2024) looks at the single outcome first and then extend to a single data block.

Calculating the KL divergence over the set of distributions $\mathcal{H}$ are convex while the calculation of KL over parameter space Θ is **ALSO** convex??

> note that we cannot draw this conclusion if $\mathcal{H}'_0$ is not convex; for then the distribution p'_{W° may not correspond to the distribution p_{W° in the original problem — this correspondence is only guaranteed if p'_{W° coincides with some p'_θ.

The part I am most confused about is the relationship between the convexity of the hypothesis space (original and modified) and parameter space. What we want is a $\mathcal{H}'_0$ that is compact and convex.

Proof: Convexity of the Kullback-Leibler divergence, it gives the proof of convexity in the pair of probability distributions (p, q) in discrete cases but the same could be applied to continuous cases with integration.

5.3.6. *General case paper discussion with Peter.* If we have 1-d regular exponential family with the same sufficient statistics, let's assume we are talking about Turner's example.

The GRO one is the conditional E-variable and putting the prior on the curve (ark) results also in the RIPr one (GRO) E-variable.

Compared to this case (fixed odds ratio δ vs 1, curve vs diagonal straight line), it is not clear that if we still get the optimal E-variable if instead of a fixed δ, we put a prior on the simplex, what sort of optimality we would get if there any.

Would the conditional variable still be the optimal one?

Let $f(k) = \binom{n_a}{k}\binom{n_b}{U-k}$ and $f'(k) = \frac{n_b!}{k!(n_a-k)!(U-k)!(n_b-U+k-n_a)!} = \binom{n_a-U+k}{n_a}$

$$p_{n_a,n_b,u,\omega}(k) = \frac{\frac{n_b!}{k!(n_a-k)!(u-k)!(n_b-U+k-n_a)!}\omega^k}{\sum \frac{n_b!}{k!(n_a-k)!(u-k)!(n_b-U+k-n_a)!}\omega^j}$$

We know n_{11} is the $Y_a = k$ in our settings,

5.4. *Other misc.*

5.4.1. *Project direction on 28.07.2025.* The goal to get a conditional E-variable

- with priors on log odds ratio: Jeffrey's, conjugate, Cauchy, Laplace, Gaussian
- uniformly most powerful (UMP) test for a fixed α, ask Sebastian
- normalized maximal likelihood (NML), aligns with the MDL principle

The second amazingly achieves the most powerful test with a given test, related materials are in the reply to the safe testing but it is given in the context of unconditional one. So Peter was not sure about its look here.

Compare it with the old one (unconditional one)

- Turner's
- Universal inference (MLE)
- p-to-e calibrator ($e = 1/\sqrt{q} - 1$ maybe)

Previous results shows that the UI performs the worst, next to Turner's, a simple calibrator works best, e.g. $f(p) = \kappa p^{\kappa-1}$ for some $\kappa \in (0, 1)$ and $f(p) = p^{-1/2} - 1$. noted that the Fisher's exact test gives a very small p-value

First step is to start with a fixed prior and fixed alternative.

5.4.2. *Re-orientate at 01.04.2025.* The biggest reason that I have been so disoriented is that I have so many documents, tests scattered across platforms and places: GitHub issue, Personal repo, Forked repo. Originally, I wanna use Issue as a tracking board, personal repo for experiments, and the forked repo for dev. In the end, I became over-whammed by all and often found myself wasting energy, organising thoughts even after just 1 day.

Thus, I have decided that I will put everything related to 2x2 in this LaTeX document. I will still use my Personal repo for testing but I will not use GitHub issue any more. I will write exclusively here as I find writing with LaTeX is still the simplest when even a little bit of math is involved. I will use the item or numbered list here and ~~strike through it~~ upon completion or delete it when it is not longer relevant.

6. Reverse Inverse Projections (RIPr).

6.1. *numeraire.* It discuss a point alternative Q against a composite null $\mathcal{Q}$ with basically no assumptions. For simplicity, use Q for a point alternative and $\mathcal{P}$ for composite null [1]

THEOREM 6.1 (Li). *$\mathcal{P}$ is a convex set of probability measures (*composite null *in testing), and Q a measure on Ω (*point alternative*). If $\mathrm{D_{KL}}(Q \,\|\, P) < \infty$ for all $P \in \mathcal{P}$, then there exist a unique probability measure P^* such that*

1. $\ln p_n \to \ln p^*$ *in* $L_1(Q)$
2. $\int_\Omega \ln \frac{\mathrm{d}Q}{\mathrm{d}P^*}\,\mathrm{d}Q = \mathrm{D_{KL}}(Q \,\|\, \mathcal{P})$
3. $\int_\Omega \frac{\mathrm{d}Q}{\mathrm{d}P^*}\,\mathrm{d}P \leq 1$ *for all* $P \in \mathcal{P}$

We consider a description gain given by

$$\mathrm{D_{KL}}(Q \,\|\, P \rightsquigarrow P') = \int_\Omega \ln\left(\frac{\mathrm{d}P'}{\mathrm{d}P}\right)\mathrm{d}Q - (P'(\Omega) - P(\Omega))$$

For a set $\mathcal{P}$, *maximum gain from Q to any measure in $\mathcal{P}$*

$$\mathrm{D_{KL}}(Q \,\|\, P \rightsquigarrow \mathcal{P}) = \sup_{P' \in \mathcal{P}} \mathrm{D_{KL}}(Q \,\|\, P \rightsquigarrow P')$$

THEOREM 6.2 (Tyron). *Suppose* $\inf_{P\in\mathcal{P}} \mathrm{D_{KL}}(Q \,\|\, P \rightsquigarrow \mathcal{P}) < \infty$*, then there exists a measure P^* in $\mathcal{P}$ such that for every sequence P_n in $\mathcal{P}$ we have*
$\mathrm{D_{KL}}(Q \,\|\, P_n \rightsquigarrow \mathcal{P}) \to \inf_{P\in\mathcal{P}} \mathrm{D_{KL}}(Q \,\|\, P \rightsquigarrow \mathcal{P})$ *as* $n \to \infty$

- $p_n \to p^*$ *in some complete space, not just* $\ln$ *convergence in* $L_1(Q)$
- .....
- *For any* $P \in \mathcal{P}$,

$$\int_\Omega \frac{\mathrm{d}Q}{\mathrm{d}P^*}\,\mathrm{d}P \leq Q(\Omega) + P(\Omega) - \liminf_{n\to\infty} P_n(\Omega)$$

Difference between Tyron and Li

- I could not see the extended part where Li's fails, where the divergence is infinity.

THEOREM 6.3 (Numeraire).

6.2. *Li's Algorithm.* Originally developed by **?**, the following algorithm is written based on the notes from Grünwald, de Heide and Koolen (2024a); **?**. To quote **?**, Li's inequality requires the family $\mathcal{Q}$ to have a uniformly bounded density ratio.

Li obtains the RIPr in a greedy manner where the KL divergence between distribution Q onto the convex hull of a set of distributions $\mathcal{Q}$ (composite null) is minimized. It is assumed that the KL divergence between Q and any distribution $Q \in \mathcal{Q}$ is finite [2] (often call "nondegenerate" condition).

Here, the distribution Q' and α is chosen (coupled) such that the divergence is minimized. The minimizer need not be unique.

Regularity condition on alternative

---

[1] Tyron use P and $\mathcal{C}$ for point alt. and composite null, so was Li

[2] Which should I use? $\mathcal{Q}$ or $\mathbb{Q}$

Algorithm 1: Li's Algorithm

1. $Q_{(1)} = \arg\min_{Q \in \mathcal{Q}} D(P \| Q)$
2. **for** $m = 2, 3, \ldots, K$ **do**
3. $\quad Q := \alpha Q_{(m-1)} + (1-\alpha) Q'$
4. $\quad \alpha, Q' \leftarrow \arg\min_{\alpha, Q'} D(P \| Q)$

Li's algorithm is apparently greedy with high fluctuation in initial steps. Additionally, this task is computationally expensive, and it is not clear of the convexity.

Is the returned mixture in the convex hull?

The returned $Q_{(m)}$ is in the convex hull. The first step returned a single element in $\mathcal{Q}$ with the smallest KL divergence. Iteratively, the linear combination is still in the convex hull, i.e.

$$
\begin{aligned}
Q_{(2)} &= \alpha_1 \cdot Q_{(1)} + (1-\alpha_1) \cdot Q'_{(1)} \\
Q_{(2)} &= \alpha_2 \cdot Q_{(2)} + (1-\alpha_2) \cdot Q'_{(2)} \\
&= \alpha_2 \cdot \alpha_1 \cdot Q_{(1)} + \alpha_2 \cdot (1-\alpha_1) \cdot Q'_{(1)} + (1-\alpha_2) \cdot Q'_{(2)}.
\end{aligned}
$$

It is clear that $Q_{(2)}$ or $Q_{(m)}$ would still be a convex combination of elements in $\mathcal{Q}$.

7. Nonnegative martingales and E-process.

7.1. *Practical.* First of all, how do one show a process is a e-process? Let's assume, we *propose* a potential random process S_t

Definition 7.1 (inadmissible). For some $t \in \mathbb{N}_0$, and some distribution $P \in \mathcal{P}$, there exist $P(p_t' < p_t) > 0$.

The argument is the same for E-process with 'greater' rather than 'smaller'
Gap between necessary and sufficient conditions

Example 7.2. Let U be a uniform random variable in $[0, 1]$ and $\mathcal{Q}$ is the set of probability measures where X_1 is Bernoulli and $X_2, X_3, \cdots = 0$. Let thje

7.2. *Stuff.* Being integrable or not, I think, is the main contribution from **?**.

Question 7.3. Which class of random processes is bigger? E-processes or martingales?

If the e-process is defined by the stuff upper bounded NNSM, then martingales is a bigger? Or should I say for a set of distribution $\mathcal{Q}$ we can always find some NNSM upper bounding the e_t??

7.3. *Notation.* We limit ourself to discrete time only $n \in \mathbb{N}_0 = \{0, 1, 2 \dots\}$, usually written as $n \geq 0$. Fix a probability triplet $(\Omega, \mathcal{A}, \mathcal{P})$ and $\mathcal{F}_t$, a sigma-field at time t, defines a filtration, $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \mathcal{A}$. We call $\mathcal{F}_t$ the canonical filtration and $X = \{X_n\}_{n \geq 0}$ a stochastic process adapted to $\mathcal{F}$ if X_n is $\mathcal{F}_n$-measurable.

Definition 7.4. An E-process for a family $\mathcal{P}$ of probability measures, namely $\mathcal{P}$-E-process, is a nonnegative process E, adapted to some filtrations such that

$$\mathbb{E}_{\mathtt{P}}\left[E_\tau\right] \leq 1 \quad \text{for every stopping time} \quad \tau$$

It can also be defined as any nonnegative process that is upper bounded, for every $P \in \mathcal{P}$, by a P-martingale starting at one.
Focus on the section 5-8 in **?**.

Definition 7.5 (Admissibiliy).

Have to check Prop. 36 first where the set Q is reduced to simpler subclass: 1) null with Lebesgue densities up to finite time 2) discrete distributions with iid.
Lemma 38: subGaussian with one atom. We can always create a subGaussian RV that has point mass in some ball (a, b) while behaving like Gaussian outside

8. Integrals. What is it other than the area under curve?

Interesting examples where the Riemann's integration fails is that $f(x) = 1$ if x is rational and 0 otherwise over the interval $[0,1]$. Why this function's integral is problematic in Riemann's definition.

Lebesgue used measure theory to define integral.

> Many times, when Riemann fails, Lebesgue works.

Set being countable and uncountable rely on if we can arrange them in a 'readable' sequence, could be infinite like rational numbers.

Proving that $\mathbb{R}$ is uncountable is not so trivial: decimal expansion.

No matter how hard you try to write the numbers in $[0,1)$, there is always some values left out.

Continuum hypothesis: How do you check which ∞ is bigger? Or which set has the higher cardinality?

Definition 1.2.2 ()

EXAMPLE 8.1 (Set of Lebesgue measure zero, 1.2.2). A set S is zero in Lebesgue measure when it can be covered with a sequence of open intervals $(I_1, I_2, \dots)$. And the sum $\sum_0^\infty m(I_n)$ can be made arbitrarily small

In other words, for any $\epsilon > 0$, we can always find a sequence of I_n that covers S.

THEOREM 8.2. *Any countable infinite set of S has Lebesgue measure zero.*

REMARK 8.3. The set of measure zero would not change a damn thing on f

8.1. *Filtrations and sigma-fields.*

DEFINITION 8.4 (Filtration). Let $(\Omega, \mathcal{A}, P)$ be a probability space. An increasing sequence $\mathcal{F}_n$ of sub-σ algebras of $\mathcal{A}$ (i.e. $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \mathcal{F}_n \subseteq \cdots \subseteq \mathcal{A}$) is called a filtration.

The sub-σ-algebra is just the sigma algebra of the X_0, X_1, X_n.

What is the difference between the stopping time and random times that can take possibly infinity.

Filtration sigma-field at time t, filtration is a n increasing sequence of sigma-fields.

What is the lim sup here? A_t is an adapted sequences of events in some filtered probability space.

$$A_\infty := \limsup_{t\to\infty} A_t := \bigcap_{t\in\mathbb{N}} \bigcup_{s\geq t} A_s$$

DEFINITION 8.5 (Absolute continuous). Let p, q be two probability distributions. p is called *absolutely continuous* with respect to q if the

More explicitly, this reads ($p \ll q$) that

$$q(A) = 0 \quad \Rightarrow \quad p(A) = 0, \quad \text{for any } A \in \mathcal{F}_t \text{ and } t \in \mathbb{N}.$$

What about *CORASENING*?

Before writing this section, I sometimes came across expressions such as $p(x) = A\,\mathrm{d}x$ or $p(x) = A'm(\mathrm{d}x)$ in defining exponential family. It relates to the fundamental concept of measure and a density in measure theory. When I see $\mathrm{d}x$ in evaluating integral, e.g. $f(x) = \int_{\mathbb{R}} x\,\mathrm{d}x = x^2$, it usually refers to the Lebesgue measure. In the case of $m(\mathrm{d}x)$, it generally emphasize that you are integrating with respect to a general measure m.

We skipped defining the carrier density and its measure for now.

- Correct carrying density
- Check if cumulant function can be differentiated infinitely

9. Exponential Family. The exponential family is a collection of parametric models with elegant properties. Peter and Hao utilised it to arrive some nice results about the optimality of E-variables, mainly relying on the duality. We called a model belonging to the exponential family if the underlying distributions can be written below.

Definition 9.1 (Exponential Family).

$$p_\beta(U) := \exp\left(\beta^T t(U) - \psi(\beta)\right) r(U) \tag{9.1}$$

$$= \frac{1}{Z(\beta)} \exp\left(\beta^T X\right) r(U). \tag{9.2}$$

- U: *random variable*
- $X = t(U)$: *sufficient statistics*
- β: *canonical parameter*
- $\psi(\beta)$: *cumulant function*
- $r(y)$: the *carrying density*
- $Z(\beta)$: the *partition function*, $\psi(\beta) = \log Z(\beta)$

9.1. *Basic Properties.* Cumulant generating function $\psi(\beta)$: The first and second *cumulant* is mean and variance. $\psi(\beta)$ is differentiable infinitely.

Definition 9.2 (Minimal). We say an exponential family representation is *minimal*. If not minimal, there is t_k that can be represented by the linear combination of t_j for $j \neq k$, requiring 1 less parameter.

We usually assume minimal if not mentioned.

Definition 9.3 (Canonical Parameter Space). Canonical parameter space $\mathtt{B}$ is the set of parameter β where the following integration is finite

$$\mathtt{B} := \left\{ \beta : \int_{\mathcal{U}} \exp\left(\beta^T t(U)\right) r(U) < \infty \right\}.$$

Remark 9.4. Every exponential family has a canonical parameterization with carrier density p_0 such that the canonical parameter space contains the origin, i.e. $\mathbf{0} \in \mathtt{B}$. (Section 18.4.3 in MDL)

Proof. If $\mathbf{0} \notin \mathtt{B}$ or $r(U)$ is not a carrier density, we can pick any $\beta_0 \in \mathtt{B}$ and set

$$r'(U) = \frac{r(U)\exp(\beta t(U))}{\sum_U r(U)\exp(\beta t(U))}$$

Now the family corresponding to r' with parameter $\beta' = \beta - \beta_0$ is identical to the original family containing $\mathbf{0}$. $\square$

DEFINITION 9.5 (Regular Exponential Family). If the canonical parameter space of exponential families are nonempty open set, we call such families *regular*.

DEFINITION 9.6 (Full Exponential Family). The family is called *full* if the dimension of β equals the dimension of $\mathtt{B}$.

DEFINITION 9.7 (Order). The order of an exponential family is the minimal dimension of $t(u)$ such that we can express the family using Eq. (9.1).

DEFINITION 9.8 (Minimal Exponential Family). An exponential family is referred to as *minimal* if: a) there are no linear constraints among the components of the parameter vector β; b) there are no linear constraints among the components of the sufficient statistic $t(u)$ (in the latter case, with probability one under the measure m).

EXAMPLE 9.9 (Non-minimal distribution). The simplest would be a multinomial distribution with parameter $()$. The PMF can be written as bit

$$s$$

We can reparameterize the probability distribution even in the case of minimal distribution

add example

. Given a (?arbitrary) set Θ and a mapping $\Phi : \Theta \to \mathtt{B}$, we consider the densities with canonical parameters replaced by $\Phi(\theta)$

$$p_\theta(u) := \exp\big(\Phi(\theta)^T t(u) - \phi(\Psi(\theta))\big).$$

Here Φ is a one-to-one mapping whose image is all of $\mathtt{B}$.

If Φ's image is a strict subset of $\mathtt{B}$ ($\Phi(\Theta) \subset \mathtt{B}$), it is then OK to reparameterize on that subset. If it can not be reducible, we refer this exponential family as curved.

We are also interested in cases in which the image of ψ is a strict subset of N. If this subset is a linear subset, then it is possible to transform the representation into an exponential family on that subset. When the representation is not reducible in this way, we refer to the exponential family as a curved exponential family.

DEFINITION 9.10 (Curved Exponential Family). TBH

EXAMPLE 9.11 (Normal distribution with variance equalling to mean). Let $u \in \mathbb{R} \sim \mathcal{N}(\mu, \mu^2)$, $\mu \neq 0$. The density of u is

$$\begin{aligned}
p_\mu(u) &= |\mu|^{-1}(2\pi)^{-1/2}\exp\left(-\frac{1}{2}(\frac{u-\mu}{\mu})^2\right) \\
&= (2\pi)^{-1/2}\exp\left(-\frac{u^2}{2\mu^2} + \frac{u}{\mu} - \frac{1}{2} + \log(|\mu|)\right) \\
&= \exp\big(\beta^T t(u) - \psi(\beta)\big).
\end{aligned}$$

The $\beta^T = (-1/2\mu^2, 1/\mu)$, the sufficient statistics $t(u) = (u^2, u)$. The dimension of the sufficient statistic is more than the dimension of β for curved exp. family.

I have found some other versions of the same statement regarding minimal exponential family:

- A minimal exponential family is where the $t(u)$ are linearly independent
- A minimal exponential family is one where representation reaches the *order*

All share the same statement regarding the linear dependency of sufficient statistics $t(u)$, only Michael I. Jordan's Chapter 8 stated on β. I wonder if linear dependency in β implies linear dependency in $t(u)$, i.e.

$$a^T\beta = C \Leftrightarrow b^T t(u) = C'$$

He also claimed that non-minimal families can always be reduced to minimal families via a suitable transformation and reparameterization.

If an exponential family is not minimal, it is called *overcomplete*. Both minimal and overcomplete representations are useful

9.2. *Mean-value parameterization.*

Definition 9.12 (Matching pair). Let $\mathcal{P}, \mathcal{Q}$ be exponential family for random variable U. We can they are *matching pair* if

- they are mutually absolutely continuous wrt each other
- they have the same sufficient statistics X
- their mean-value space, $\mathtt{M}_q \subseteq \mathtt{M}_p$
- for every matching canonical parameterisation of $\mathcal{P}$ and $\mathcal{Q}$, we have $\mathtt{B}_p \subseteq \mathtt{B}_q$

9.3. *Simple Case.*

- Why I think this is the carrier probability with some measure v $p_{\boldsymbol{\mu}^*} = p_{\mathbf{0},\boldsymbol{\mu}^*} \quad q_{\mathbf{0},\boldsymbol{\mu}^*} = q = q_{\boldsymbol{\mu}^*}$:

  It is rather simple when you write down $\boldsymbol{\beta} = \mathbf{0}$

$$\begin{aligned} p_{\mathbf{0};\boldsymbol{\mu}^*}(u) &:= \frac{1}{Z_p(\mathbf{0};\boldsymbol{\mu}^*)} \exp\big(\mathbf{0}^T t(u)\big) \cdot p_{\boldsymbol{\mu}^*}(u) \\ &= \frac{p_{\boldsymbol{\mu}^*}(u)}{Z_p(\mathbf{0};\boldsymbol{\mu}^*)} \\ &= \frac{p_{\boldsymbol{\mu}^*}(u)}{\int \exp(\mathbf{0}^T t(u)) p_{\boldsymbol{\mu}^*}(u)\, \mathrm{d}\nu} = p_{\boldsymbol{\mu}^*}(u) \end{aligned}$$

- A naive question follows: what is the distribution $p_{\boldsymbol{\mu}^*}(u)$? Is it in the mean-value parameterization or canonical form?

9.4. *General Case.*

- Why Hao explicitly denote the mean of the carrier density in Grünwald, de Heide and Koolen (2024a):

We first discuss the Gaussian location family where the mean $\boldsymbol{\mu}^* \in \mathtt{M}_p = \mathbb{R}^d$. The null distribution will be family with $\boldsymbol{\mu}^*$ and a positive semidefinite covariance matrix Σ_p and the alternative $\mathcal{Q} = \{Q\}$ would be Gaussian distribution with the same $\boldsymbol{mu}^*$ and a covariance matrix $\Sigma_p \neq \Sigma_q$.

The notation D_{Gauss} in Eq. (3.2.1) in **?** is just the KL divergence. Here $X = U = (X_1, \ldots, X_d)$ is the d-dimensional random vector with distribution p or q

$$
\begin{aligned}
D_{\text{GAUSS}}(B) &= \mathrm{D}_{\mathrm{KL}}(P \,\|\, Q) \\
&:= \int_{\mathcal{U}} p(X) \log \frac{p(X)}{q(X)} = \mathbb{E}_p \left[\log \frac{p(X)}{q(X)} \right] \\
&= \mathbb{E}_p \left[\log \frac{(2\pi)^{-d/2} \det(\Sigma_p)^{-1/2} \exp\left(-\frac{1}{2}(X-\boldsymbol{\mu}^*)^T \Sigma_p^{-1} (X-\boldsymbol{\mu}^*)\right)}{(2\pi)^{-d/2} \det(\Sigma_q)^{-1/2} \exp\left(-\frac{1}{2}(X-\boldsymbol{\mu}^*)^\top \Sigma_q^{-1} (X-\boldsymbol{\mu}^*)\right)} \right] \\
&= \mathbb{E}_p \left[\log \frac{\det(\Sigma_q)^{-1/2} \exp\left(\Sigma_p^{-1}\right)}{\det(\Sigma_q)^{-1/2} \exp\left(\Sigma_q^{-1}\right)} \right] \quad \text{Wrong!}
\end{aligned}
$$

For more details, check https://statproofbook.github.io/P/mvn-kl.html for cases where the means are different.

Why Eq. (3.4.2) holds? The part I don't understand is that why putting a prior will equal to $p_{\boldsymbol{\mu}^*}$

$$
S_{\text{COND}} := \frac{q_{W_1}(U^{(n)} \mid Z)}{p_{W_0}(U^{(n)} \mid Z)} \overset{?}{=} \frac{q_{\boldsymbol{\mu}^*}(U^{(n)} \mid Z)}{p_{\boldsymbol{\mu}^*}(U^{(n)} \mid Z)}
$$

The key is just simple derivation with the fact that $\hat{X}_{|n} := \sum X_i / n \sim \mathcal{N}(n\boldsymbol{\mu}^*, n\Sigma_q)$. Setting $Z = \hat{X}_{|n} / \sqrt{n} \sim \mathcal{N}(\sqrt{n}\boldsymbol{\mu}^*, \Sigma_q)$, we have

$$
\begin{aligned}
q_{W_1}(X^{(n)} \mid Z) &:= \frac{q_{W_1}(X^{(n)}, Z)}{q_{W_1}(Z)} = \frac{q_{W_1}(X^{(n)})}{q_{W_1}(Z)} \overset{iid}{=} \frac{\prod_{i=1}^n q_{W_1}(X_i)}{q_{W_1}(Z)} \\
&= \frac{\prod (2\pi)^{-d/2} (\det \Sigma_q)^{-1} \exp\left(-1/2 (X_i - \boldsymbol{\mu}^*)^\top \Sigma_q^{-1} (X_i - \boldsymbol{\mu}^*)\right)}{(2\pi)^{-d/2} (\det \Sigma_q)^{-1} \exp\left(-1/2 (Z - \sqrt{n}\boldsymbol{\mu}^*)^\top \Sigma_q^{-1} (Z - \sqrt{n}\boldsymbol{\mu}^*)\right)} \\
&= \frac{\exp\left(\sum_{i=1}^n X_i^\top \Sigma_q^{-1} X_i - 2\sqrt{n}\boldsymbol{\mu}^{*\top} \Sigma_q^{-1} Z + n\boldsymbol{\mu}^{*\top} \Sigma_q^{-1} \boldsymbol{\mu}^*\right)}{\exp\left(Z^\top \Sigma_q^{-1} Z - 2\sqrt{n}\boldsymbol{\mu}^{*\top} \Sigma_q^{-1} Z + n\boldsymbol{\mu}^{*\top} \Sigma_q^{-1} \boldsymbol{\mu}^*\right)} \\
&= \exp\left(\sum_{i=1}^n (X_i - Z)^\top \Sigma_q^{-1} (X_i - Z) \right)
\end{aligned}
$$

The equality follows due to the joint distribution $X^{(n)}$ and Z is simply $X^{(n)}$.

For UI case, I don't see any sign of splitting? It looks like just ML prequential settings

What makes the E-process-ness? th

Why the $S_{\text{SEQ}} = S_{\text{SEQ,RIP}}$ in Thm. 2?

9.5. *One-dimensional special case.*

$$
\begin{aligned}
&\mathcal{G} := \{g_\eta(y), \eta \in A, \in \mathcal{Y}\}, \quad A \text{ and } \mathcal{Y} \in \mathbb{R}^p \\
&g_\eta(y) := \exp\left(\eta y - \psi(\eta)\right) g_0(y) m(\mathrm{d}y)
\end{aligned}
$$

REMARK 9.13 (Carrying density in p.3 efron). Any $g_{\eta_0}(y) \in \mathcal{G}$ could be the carrier density and the members of $\mathcal{G}$ are absolutely continuos with respect to each other, i.e. their null measure sets agree.

REMARK 9.14. *Cumulant generating function* for $\psi(\eta)$ originates from the old techniques for finding expectations, variances and higher-order moments.

9.6. *Moment Relationships.* We can differentiate $\exp(\psi(\eta)) = \int_{\mathcal{Y}} \exp(\eta y) g_0(y) m(\mathrm{d}y)$

$$\dot{\psi}(\eta) \exp(\psi(\eta)) = \int_{\mathcal{Y}} y \exp(\eta y) g_0(y) m(\mathrm{d}y)$$

$$\mathbb{E}_{g_0}[y] = \dot{\psi}(\eta) = \int_{\mathcal{Y}} y \exp(\eta y - \psi(\eta)) g_0(y) m(\mathrm{d}y)$$

Differentiating $\exp(\psi(\eta))$ twice gives:

$$(\ddot{\psi}(\eta) + \dot{\psi}(\eta)) \exp(\psi(\eta)) = \int_{\mathcal{Y}} y^2 \exp(\eta y) g_0(y) m(\mathrm{d}y)$$

$$\mathrm{Var}(y) = \ddot{\psi}(\eta) = \int_{\mathcal{Y}} (y^2 - y) \exp(\eta y - \psi(\eta)) g_0(y) m(\mathrm{d}y)$$

Efron's book (Chap. 2)

$$\mathcal{G} := \{g_\eta(y), \eta \in A, y \in \mathcal{Y}\}, \quad A \text{ and } \mathcal{Y} \in \mathbb{R}^p$$

$$g_\eta(y) := \exp\left(\eta^T y - \psi(\eta)\right) g_0(y) m(\mathrm{d}y)$$

- $g_0(y)$: *carrying density* w.r.t. some *carrying measure* $m(\mathrm{d}y)$ on $\mathcal{Y}$
- A is the *canonical parameter space*:
- $\psi(\eta)$ is the *normalizing function* or *cumulant generating function* (CGF)

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